



Statistical Reasoning 3

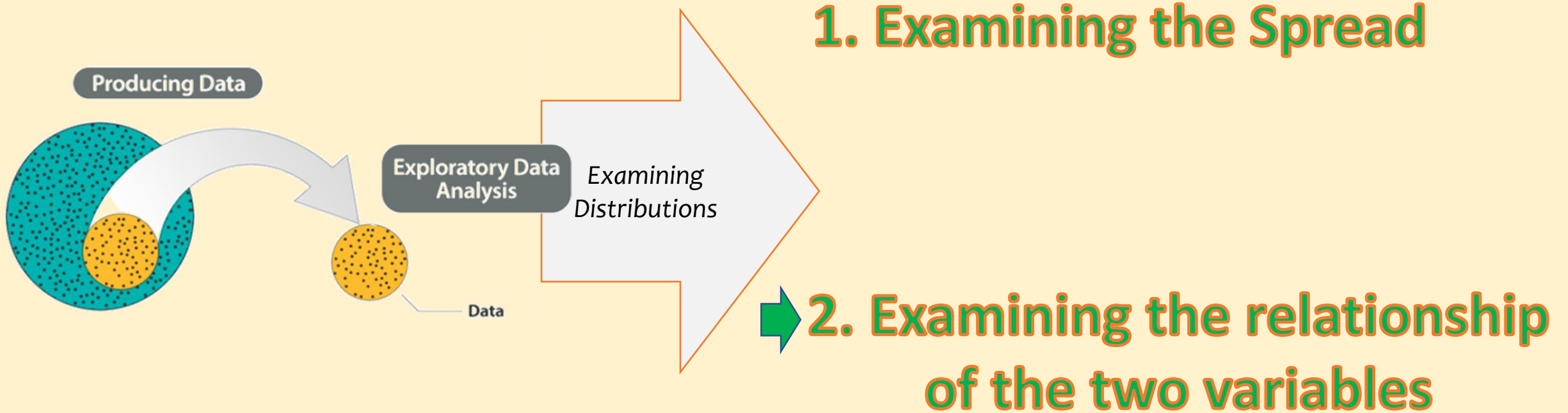
PLAN FOR TODAY!

- ☐ Examining Relationships
- ☐ Exercises
- ☐ Examining Relationships cont.

Announcement

- The prompt for your semester project is online. Please familiarize yourself with the prompt.
 - **Deliverable 1:** March 24th, 11:59 pm
 - **Deliverable 2:** April 5th, 11:59 pm
 - **Deliverable 3:** April 19th, 11:59 pm
 - **Deliverable 4:** May 1st, 11:59 pm

Examining Relationships



Examining Relationships

- So far we have dealt with data obtained from one variable (either categorical or quantitative), and learned how to describe the distribution of the variable using the appropriate visual displays and numerical measures.

Examining Relationships

- Now, we will look into **two** variables at a time and explore the **relationship** between them using (as before) visual displays and numerical summaries.

❖ Examining Relationships

➡ So far we have dealt with data obtained from one variable (either categorical or quantitative), and learned how to describe the distribution of the single variable using the appropriate **visual displays** and **numerical measures**.

Now, we examine the **relationships** by considering and looking at two variables at a time – through **visual displays and numerical summaries**. In most studies involving two variables, each of the variables has a role:

- The **explanatory variable** (also commonly referred to as the independent variable) – the variable that claims to explain, predict, or affect the response; and
- The **response variable** (also commonly referred to as the dependent variable) – the outcome of the study.
- *Example - We want to explore whether the outcome of the study – the score on a test – is affected by the test-taker's gender. Therefore, **Gender is the explanatory variable** and **Test score is the response variable***

Single Variable

Example - Commute time of HU students to campus



2 Variables

Example - Commute time & academic performance

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

Exploratory Data Analysis

Examining Relationships - from one variable (either categorical or quantitative) to examining relationships, by looking at two variables at a time – to explore the relationship between them using visual displays and numerical summaries.

- In most studies involving two variables, each of the variables has a role. We distinguish between:
 - The **explanatory variable** (also commonly referred to as the independent variable) – **the variable that claims to explain, predict, or affect the response**; and
 - The **response variable** (also commonly referred to as the dependent variable) – **the outcome of the study**.
 - Example - We want to explore whether the outcome of the study – the score on a test – is affected by the test-taker's gender. Therefore, **Gender is the explanatory variable** and **Test score is the response variable**

Exploratory Data Analysis

Examining Relationships - from one variable (either categorical or quantitative) to examining relationships, by looking at two variables at a time – to explore the relationship between them using visual displays and numerical summaries.

- In most studies involving two variables, each of the variables has a role. We distinguish between:
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 - The **response variable** (also commonly referred to as the dependent variable) – **the outcome of the study**.
 - Example - We want to explore whether the outcome of the study – the score on a test – is affected by the test-taker's gender. Therefore, **Gender is the explanatory variable** and **Test score is the response variable**

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

If we further classify each of the two relevant variables according to type (categorical or quantitative), we get the following four possibilities for **role-type classification**:

1. **Categorical explanatory and quantitative response**
2. **Categorical explanatory and categorical response**
3. **Quantitative explanatory and quantitative response**
4. **Quantitative explanatory and categorical response**

Exploratory Data Analysis

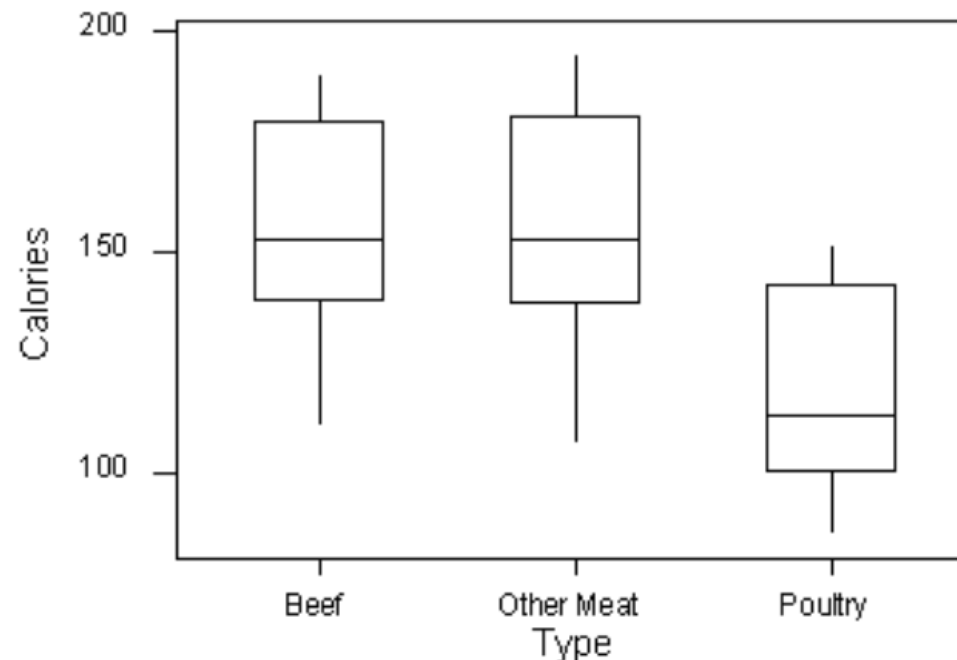
1

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

Explanatory

Response

	Type	Calories
Brand 1	Beef	186
Brand 2	Poultry	129
Brand 3	Beef	181
Brand 4	Meat	173
.	.	.
.	.	.
.	.	.
Brand 54	Poultry	144



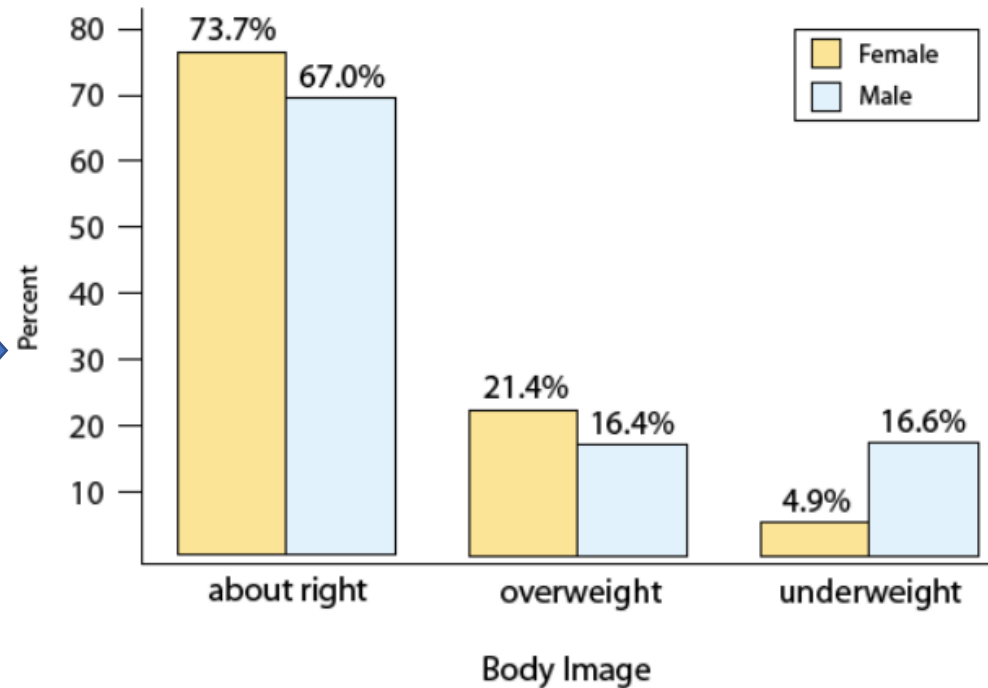
Exploratory Data Analysis

2

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	73.7%	21.4%	4.9%	100%
	Male	67.0%	16.4%	16.6%	100%



EXERCISES

16. In example 2 (see corresponding module reading), the number of calories is the { explanatory, response } variable, and the type of hot dog is the { explanatory, response } variable.

✍ 16. In example 2 (see corresponding module reading), the number of calories is the { explanatory, response } variable, and the type of hot dog is the { explanatory, response } variable.

- In this study we are exploring whether the type of hot dog affects the number of calories it has. Thus, the number of calories is the response variable.
- Thus, the type of hot dog is the explanatory variable.
- The key to deciding which variable is the explanatory variable and which is the response variable is to decide which variable affects (or explains) the other. If we can say that X affects (or explains) Y , then X is the explanatory variable.
- Does the type of hot dog affect the number of calories or do the number of calories affect the type of hot dog?

✍ 17. In example 4, gender is the { explanatory, response } variable, and smoking habit is the { explanatory, response } variable.

✍ 17. In example 4, gender is the { explanatory, response } variable, and smoking habit is the { explanatory, response } variable.

- Does a person's gender affect that person's smoking habits, or does a person's smoking habits affect that person's gender?
- Note that in most studies involving gender, it will be the explanatory variable. We are not trying to see how a person's smoking habits affect the person's gender, but rather explore whether gender is one of the factors that can explain whether the person is a smoker or not.
- If a person's gender is one of the factors that can explain whether the person is a smoker or not, then smoking habits is the response variable

 18. In example 6, age is the { explanatory, response } variable, and legibility distance is the { explanatory, response } variable.

✍ 18. In example 6, age is the { explanatory, response } variable, and legibility distance is the { explanatory, response } variable.

Does a driver's age affect that driver's sign legibility distance or does a driver's sign legibility distance affect that driver's age?

In this study we are examining how legibility distance changes with age. Thus, we can say that age affects legibility distance, making it the explanatory variable.

Since we are examining how age affects legibility distance, legibility distance is the response variable.

✍ 19. In example 8, favorite type of music is the { explanatory, response } variable, and IQ level is the { explanatory, response } variable.

19. In example 8, favorite type of music is the { explanatory, response } variable, and IQ level is the { explanatory, response } variable.

Since the purpose of this study was to explore whether a person's IQ level can explain or help predict the person's favorite type of music, type of music is the response variable and IQ level is the explanatory variable.

20. *Example 2:*

Number of calories is the response variable and it is { categorical, quantitative }.

Type of hot dog is the explanatory variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

20. Example 2:

Number of calories is the response variable and it is { categorical, quantitative }. quantitative

Type of hot dog is the explanatory variable and it is { categorical, quantitative }. categorical

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }. $C \rightarrow Q$

21. *Example 4:*

Gender is the explanatory variable and it is { categorical, quantitative }.

Smoking habits is the response variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

21. *Example 4:*

Gender is the explanatory variable and it is { **categorical**, quantitative }. **categorical**

Smoking habits is the response variable and it is { **categorical**, quantitative }. **categorical**

Therefore this is an example of { Case $C \rightarrow Q$, **Case $C \rightarrow C$** , Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }. **$C \rightarrow C$**

 22. *Example 6:*

Age is the explanatory variable and it is { categorical, quantitative }.

Legibility distance is the response variable and it is { categorical, quantitative }

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

22. Example 6:

Age is the explanatory variable and it is { categorical, quantitative }. quantitative

Legibility distance is the response variable and it is { categorical, quantitative }. quantitative

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }. $Q \rightarrow Q$

✍ 23. *Example 8:*

Favorite type of music is the response variable and it is { categorical, quantitative }.

IQ level is the explanatory variable and it is { categorical, quantitative }.

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }.

23. Example 8:

Favorite type of music is the response variable and it is { categorical, quantitative }. categorical

IQ level is the explanatory variable and it is { categorical, quantitative }. quantitative

Therefore this is an example of { Case $C \rightarrow Q$, Case $C \rightarrow C$, Case $Q \rightarrow Q$, Case $Q \rightarrow C$ }. $Q \rightarrow C$

In each of the following three problems, you are presented with a brief description of a study involving two variables. Based on the role-type classification of the two variables, you are asked to determine which of the four cases represents the data structure of the problem.

For your convenience, here again is the role-type classification table:

 24. A store asked 250 of its customers whether or not they were satisfied with the service. The purpose of this study was to examine the relationship between the customer's satisfaction and gender.

This study is an example of which **role-type classification**?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

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This study is an example of which **role-type classification**?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.**
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

Both the explanatory (gender) and response (satisfaction) variables are categorical in this case. Therefore, this is an example of case $C \rightarrow C$.

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

In each of the following three problems, you are presented with a brief description of a study involving two variables. Based on the role-type classification of the two variables, you are asked to determine which of the four cases represents the data structure of the problem.

For your convenience, here again is the role-type classification table:

 25. A study was conducted in order to explore the relationship between the number of beers a person drinks and his/her blood alcohol content (BAC, in %).

This study is an example of which **role-type classification**?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
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In each of the following three problems, you are presented with a brief description of a study involving two variables. Based on the role-type classification of the two variables, you are asked to determine which of the four cases represents the data structure of the problem.

For your convenience, here again is the role-type classification table:

 25. A study was conducted in order to explore the relationship between the number of beers a person drinks and his/her blood alcohol content (BAC, in %).

This study is an example of which **role-type classification**?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

Both the explanatory (number of beers) and response (BAC) variables are quantitative in this case, and therefore this is an example of case $Q \rightarrow Q$.

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

In each of the following three problems, you are presented with a brief description of a study involving two variables. Based on the role-type classification of the two variables, you are asked to determine which of the four cases represents the data structure of the problem.

For your convenience, here again is the role-type classification table:

 26. A study was conducted in order to determine whether longevity (how long a person lives) is related to a person's handedness (right-handed/left-handed).

This study is an example of which **role-type classification**?

- A. Case $C \rightarrow Q$.
- B. Case $C \rightarrow C$.
- C. Case $Q \rightarrow Q$.
- D. Case $Q \rightarrow C$.

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

In each of the following three problems, you are presented with a brief description of a study involving two variables. Based on the role-type classification of the two variables, you are asked to determine which of the four cases represents the data structure of the problem.

For your convenience, here again is the role-type classification table:

26. A study was conducted in order to determine whether longevity (how long a person lives) is related to a person's handedness (right-handed/left-handed).

This study is an example of which **role-type classification**?

- A. Case **C → Q**.
- B. Case C → C.
- C. Case Q → Q.
- D. Case Q → C.

In this case, the explanatory variable (handedness) is categorical, and the response variable (longevity) is quantitative. Therefore, this is an example of C -> Q.

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – The Role-Type Classification (C → Q) - Survey of Study Habits and Attitudes (SSHA)

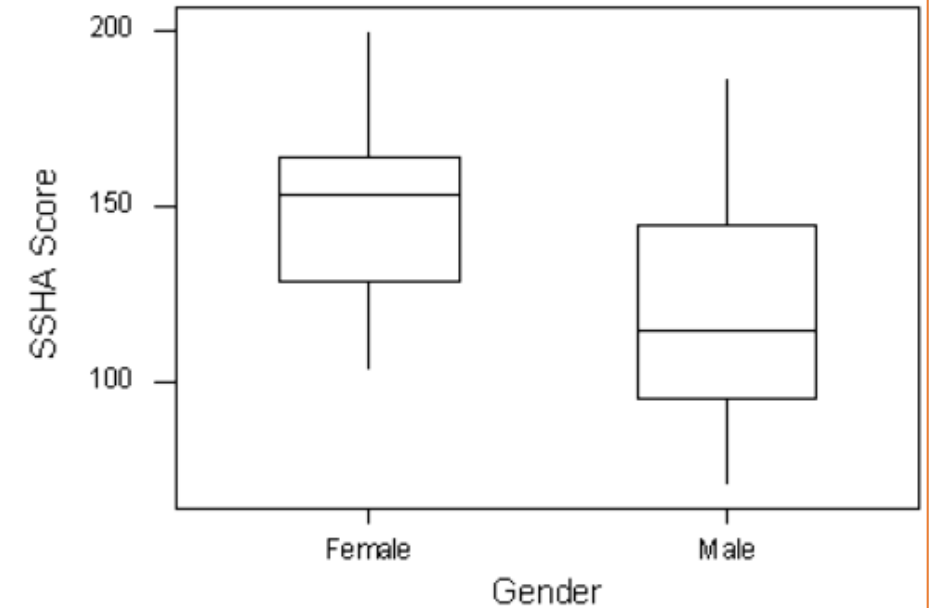
Explanatory

Response



	Gender	SSHA score
Student 1	Female	154
Student 2	Female	109
Student 3	Male	108
Student 4	Female	115
.	.	.
.	.	.
.	.	.
Student 40	Male	140

Statistic	Female	Male
min	103	70
Q1	128.75	95
Median	153	114.5
Q3	163.75	144.5
Max	200	187



Question No 26 – Do females or males have a higher **median** SSHA score?

- A. Females
- B. Males

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

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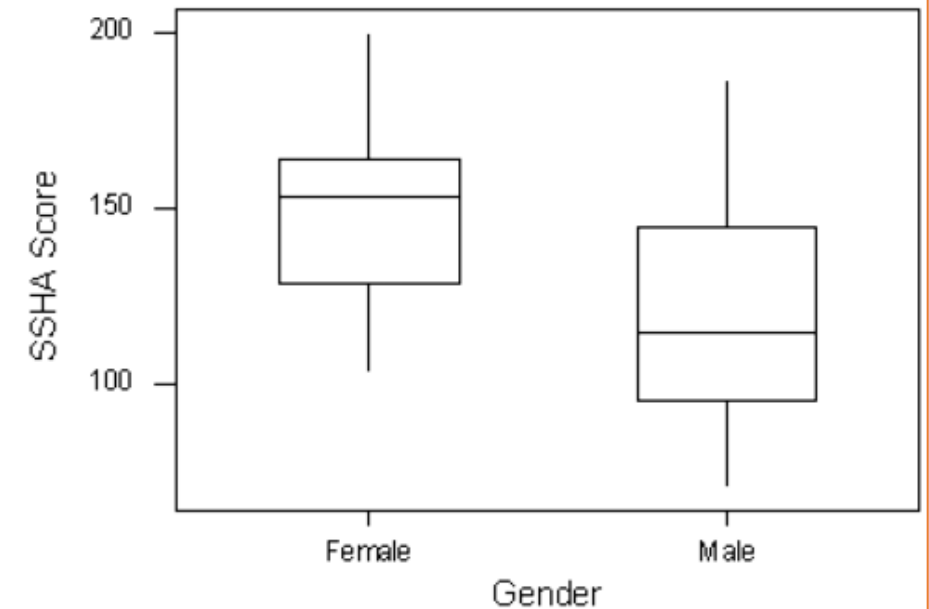
L - Reflect and discuss – The Role-Type Classification (C → Q) - Survey of Study Habits and Attitudes (SSHA)

Explanatory

Response

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Student 1	Female	154
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Max	200	187



Question No 26 – Do females or males have a higher **median** SSHA score?

A. Females

B. Males

Answer - A

The median SSHA score of females is higher than the median score for males (153 vs. 114).

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

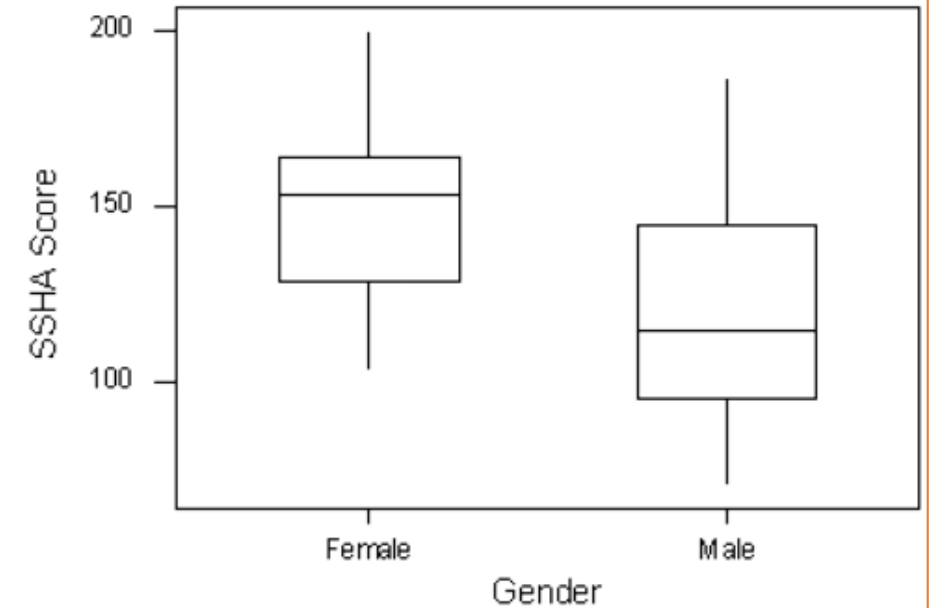
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Explanatory

Response

	Gender	SSHA score
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Question No 28 – Do females or males have higher **variability** in SSHA scores?

- A. Females
- B. Males

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

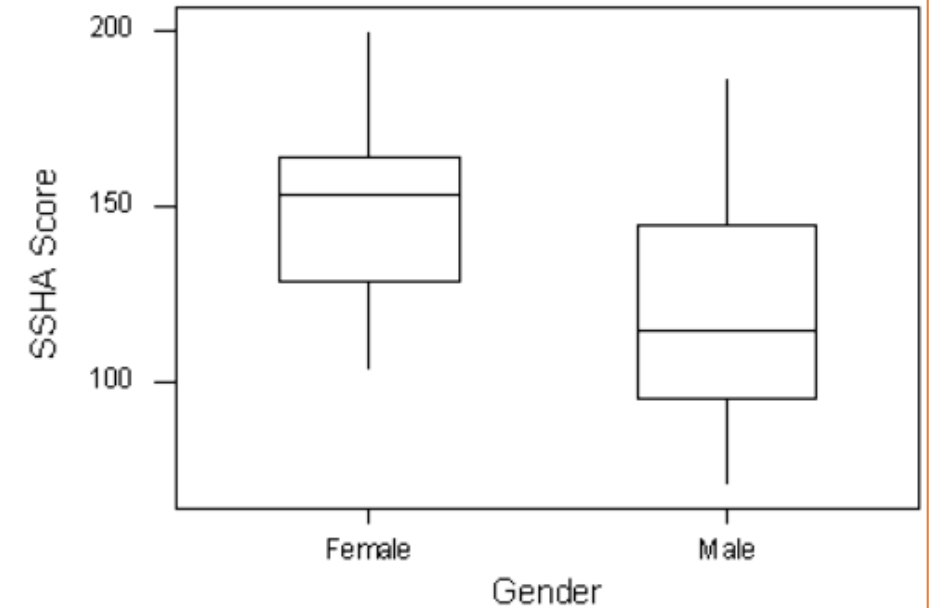
L - Reflect and discuss – The Role-Type Classification (C → Q) - Survey of Study Habits and Attitudes (SSHA)

Explanatory

Response

	Gender	SSHA score
Student 1	Female	154
Student 2	Female	109
Student 3	Male	108
Student 4	Female	115
.	.	.
.	.	.
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Student 40	Male	140

Statistic	Female	Male
min	103	70
Q1	128.75	95
Median	153	114.5
Q3	163.75	144.5
Max	200	187



Question No 28 – Do females or males have higher **variability** in SSHA scores?

- A. Females
- B. Males**

Answer - B

Males' scores display more variability both in terms of IQR (49.5 vs. 35) and in terms of the full range of scores (117 vs. 97).

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – The Role-Type Classification (C → C)

Earlier in the course, (when we discussed the distribution of a single categorical variable) we examined the data obtained when a random sample of 1,200 U.S. college students were asked about their body image (underweight, overweight, or about right.) We are now returning to this example, to address the following question:

If we had separated our sample of 1,200 U.S. college students by gender and looked at males and females separately, would we have found a similar distribution across body-image categories? More specifically, are men and women just as likely to think their weight is about right? Among those students who do not think their weight is about right, is there a difference between the genders in feelings about body image?

Explanatory		Response
Student	Gender	Body Image
.	.	.
.	.	.
student 25	M	overweight
student 26	M	about right
student 27	F	underweight
student 28	F	about right
student 29	M	about right
.	.	.
.	.	.

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – The Role-Type Classification (C → C)

Explanatory

Response



Student	Gender	Body Image
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.	.	.
student 25	M	overweight
student 26	M	about right
student 27	F	underweight
student 28	F	about right
student 29	M	about right
.	.	.
.	.	.

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

29. The number of females in our sample who feel that they are under- weight is 37
30. The number of males in our sample who feel that they are about right is 295
31. The total number of females in our sample is 760
32. The total number of students who feel they are overweight is 235

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – The Role-Type Classification (C → C)

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	$\frac{560}{760} = 73.7\%$	$\frac{163}{760} = 21.5\%$	$\frac{37}{760} = 4.9\%$	$\frac{760}{760} = 100\%$
	Male	?			100%

Question No 33 – complete the table by computing the conditional percentages for the **males**.

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – The Role-Type Classification (C → C)

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	$560/760 = 73.7\%$	$163/760 = 21.5\%$	$37/760 = 4.9\%$	$760/760 = 100\%$
	Male	?			100%

Question No 33 – complete the table by computing the conditional percentages for the **males**.

Answer

295 out of the 440 males responded “about right,” which is 67%.

72 out of the 440 males responded “overweight,” which is 16.4%.

Out of the 440 males, 73 responded “underweight,” which is 16.6%

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	73.7%	21.4%	4.9%	100%
	Male	67.0%	16.4%	16.6%	100%

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

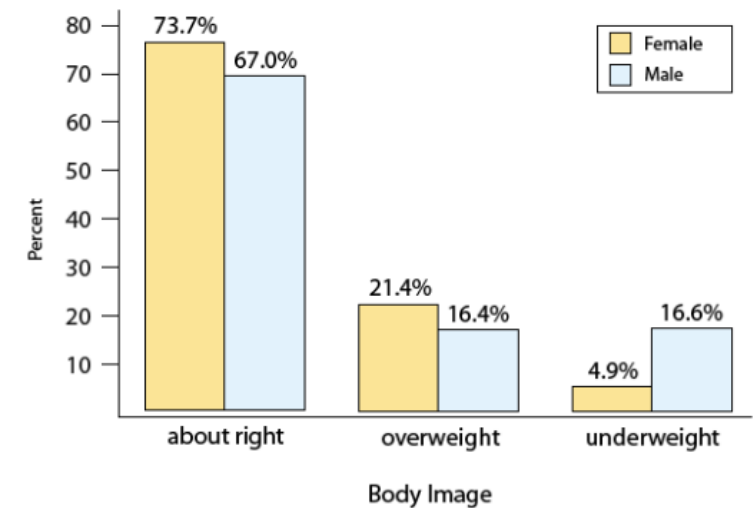
Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

O - Reflect and discuss – The Role-Type Classification (C → C)

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
	Total	855	235	110	1200

Question No 34 – The results (see double bar chart) suggest that the proportion of males who are happy with their body weight is {slightly greater, slightly less} than among female students. Females who are not happy with their body weight more often feel they are {about right, overweight, underweight}. Males who are not happy with their body weight feel they are overweight {about as often as, much more often than, much less often than} they feel they are underweight.

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	73.7%	21.4%	4.9%	100%
	Male	67.0%	16.4%	16.6%	100%



Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

O - Reflect and discuss – The Role-Type Classification (C → C)

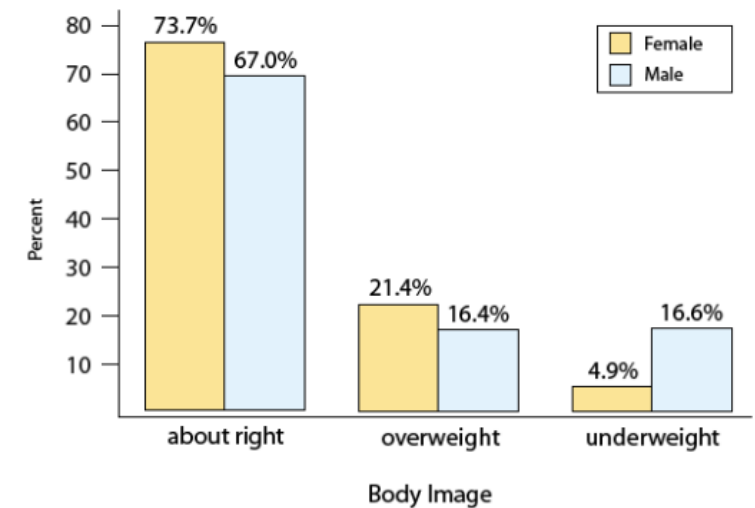
		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	560	163	37	760
	Male	295	72	73	440
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Question No 34 – The results (see double bar chart) suggest that the proportion of males who are happy with their body weight is {slightly greater, **slightly less**} than among female students. Females who are not happy with their body weight more often feel they are {about right, **overweight**, underweight}. Males who are not happy with their body weight feel they are overweight {**about as often as**, much more often than, much less often than} they feel they are underweight.

Answer

- Students who are happy with their body weight responded “about right.” 73.7% of females responded “about right,” compared to only 67% of males.
- 21.4% of females responded “overweight” compared to only 4.9% of females who responded “underweight.”
- 16.4% of males responded “overweight,” which is roughly the same as the percentage of males who responded “underweight” (16.6%).

		Body Image			
		About Right	Overweight	Underweight	Total
Gender	Female	73.7%	21.4%	4.9%	100%
	Male	67.0%	16.4%	16.6%	100%



Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

P - Reflect and discuss – The Role-Type Classification - 2

Question No 35 – Suppose a study were done to answer the question: “Is the smoking of students related to their parents’ smoking habits?” in which data were collected from 5,375 students and organized in the following two-way table:

		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	1 168	3 203	4 371
	Student Smokes	188	816	1 004
	Total	1 356	4 019	5 375

We would like to learn about the relationship between the two categorical variables “**parents’ smoking**” and “**students’ smoking**.” More specifically, since we want to explore whether parents’ smoking habits affect their children’s smoking habits, the explanatory variable is {students’ smoking, parents’ smoking} and the response variable is {students’ smoking, parents’ smoking}.

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

P - Reflect and discuss – The Role-Type Classification - 2

Question No 35 – Suppose a study were done to answer the question: “Is the smoking of students related to their parents’ smoking habits?” in which data were collected from 5,375 students and organized in the following two-way table:

		Parents’ Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	1 168	3 203	4 371
	Student Smokes	188	816	1 004
	Total	1 356	4 019	5 375

We would like to learn about the relationship between the two categorical variables “**parents’ smoking**” and “**students’ smoking**.” More specifically, since we want to explore whether parents’ smoking habits affect their children’s smoking habits, the **explanatory variable** is {students’ smoking, **parents’ smoking**} and the **response variable** is {**students’ smoking**, parents’ smoking}.

Answer -

We are exploring the impact of parents’ smoking habits on their children, and therefore, parents’ smoking is the explanatory variable and students’ smoking is the response variable.

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

P - Reflect and discuss – The Role-Type Classification - 2

Question No 36 – Based on your answer to the previous question (No 35), which of the following three tables is the appropriate table of conditional percentages with which to supplement the above contingency table “Parent’s Smoking”?

- A. Table A.
- B. Table B.
- C. Table C.

		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	1168	3203	4371
	Student Smokes	188	816	1004
	Total	1356	4019	5375

		Table A		
		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	26.7%	73.3%	100%
	Student Smokes	18.7%	81.3%	100%

		Table B	
		Parents' Smoking	
		Parents Do Not Smoke	Parents Smoke
Student Smoking	Student Does Not Smoke	86.1%	79.7%
	Student Smokes	13.9%	20.3%
	Total	100%	100%

		Table C		
		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	21.73%	59.59%	81.32%
	Student Smokes	3.50%	15.18%	18.68%
	Total	25.23%	74.77%	100%

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

P - Reflect and discuss – The Role-Type Classification - 2

Question No 36 – Based on your answer to the previous question (No 35), which of the following three tables is the appropriate table of conditional percentages with which to supplement the above contingency table “Parent’s Smoking”?

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Table A

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Table B

		Parents' Smoking	
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Student Smoking	Student Does Not Smoke	86.1%	79.7%
	Student Smokes	13.9%	20.3%
	Total	100%	100%

Table C

		Parents' Smoking		
		Parents Do Not Smoke	Parents Smoke	Total
Student Smoking	Student Does Not Smoke	21.73%	59.59%	81.32%
	Student Smokes	3.50%	15.18%	18.68%
	Total	25.23%	74.77%	100%

Answer - B

Since the explanatory variable is in the columns, we find conditional column percentages. For example, in the Parents Do Not Smoke column, we find $1,168/1,356 = 86.1\%$ and $188/1,356 = 13.9\%$.

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Q - Reflect and discuss –

The United States federal government collects information on Americans who do not have health insurance. Data from 2004 are broken down into 4 regions of the country. These data are summarized in the table provided.

Using this table, answer the following question.

Question No 37 – We wish to investigate the relationship between which two **categorical** variables?

- A. Insured and uninsured.
- B. Region and uninsured.
- C. Region and United States.
- D. Health insurance and region.
- E. Health insurance and United States.

Region	Uninsured	Insured	Total
Northeast	6,782	47,043	53,825
Midwest	7,757	57,135	64,892
South	19,090	85,800	104,890
West	11,676	55,427	67,103
Total	45,305	245,405	290,710
from 2004 census data			
figures in thousands of people			

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

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Total	45,305	245,405	290,710
from 2004 census data figures in thousands of people			

Answer - D

The entries for rows can be represented by the variable region, while the entries for the columns can be represented by the variable health insurance.

Reading Exercise – No 15 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

Q - Reflect and discuss –

The United States federal government collects information on Americans who do not have health insurance. Data from 2004 are broken down into 4 regions of the country. These data are summarized in the table provided.

Using this table, answer the following question.

Question No 38 – We wish to investigate if the region in which one lives affects whether or not one has health insurance. Which would be the **explanatory** variable?

- A. Health insurance.
- B. Region.

Region	Uninsured	Insured	Total
Northeast	6,782	47,043	53,825
Midwest	7,757	57,135	64,892
South	19,090	85,800	104,890
West	11,676	55,427	67,103
Total	45,305	245,405	290,710
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Reading Exercise – No 15 (Module | Statistical Reasoning 2)

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The United States federal government collects information on Americans who do not have health insurance. Data from 2004 are broken down into 4 regions of the country. These data are summarized in the table provided.

Using this table, answer the following question.

Question No 38 – We wish to investigate if the region in which one lives affects whether or not one has health insurance. Which would be the **explanatory** variable?

- B. Region.**

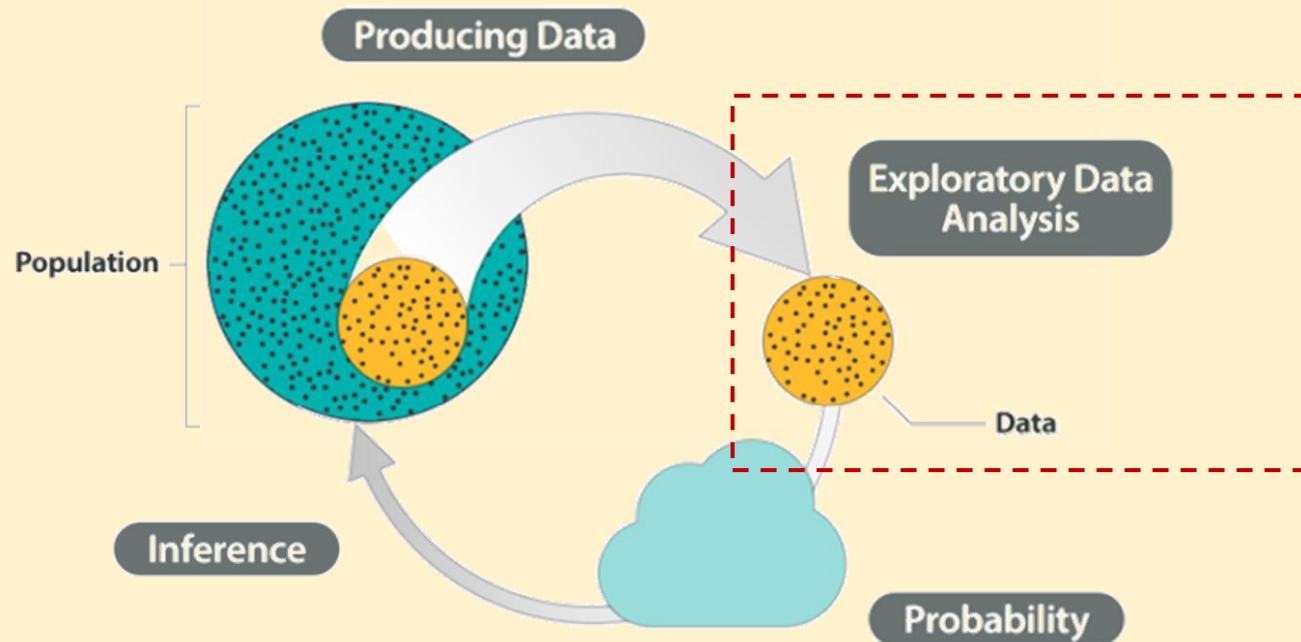
Region	Uninsured	Insured	Total
Northeast	6,782	47,043	53,825
Midwest	7,757	57,135	64,892
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from 2004 census data figures in thousands of people			

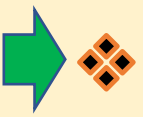
Answer - B

The explanatory variable is used to predict (explain) the response variable. We are using the region in which one lives to predict the likelihood that one has health insurance.

Reading Exercise – No 16

Examining Distributions (Examining Relationships, $Q \rightarrow Q$)





Examining Relationships

So far we have dealt with data obtained from one variable (either categorical or quantitative), and learned how to describe the distribution of the single variable using the appropriate **visual displays** and **numerical measures**.

Now, we examine the **relationships** by considering and looking at two variables at a time – through **visual displays and numerical summaries**. In most studies involving two variables, each of the variables has a role:

- The **explanatory variable** (also commonly referred to as the independent variable) – the variable that claims to explain, predict, or affect the response; and
- The **response variable** (also commonly referred to as the dependent variable) – the outcome of the study.
- Example - We want to explore whether the outcome of the study – the score on a test – is affected by the test-taker's gender. Therefore, **Gender is the explanatory variable** and **Test score is the response variable**

Single Variable

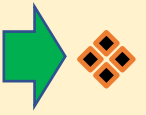
Example - Commute time of HU students to campus



2 Variables

Example - Commute time & academic performance

		Response	
		Categorical	Quantitative
Explanatory	Categorical	✓ C → C	✓ C → Q
	Quantitative	Q → C	Q → Q



Examining Relationships

2 Variables

Example - Commute time & academic performance

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

Research topic 1a

Examining the effect of commute time (HU students) to students learning motivational level (Low, medium, High)

Q → C

Research topic 1b

Examining the effect of commute time (HU students) to students GPA outcomes

Q → Q

Reading Exercise – No 16 (Module | Statistical Reasoning 2 of 4)

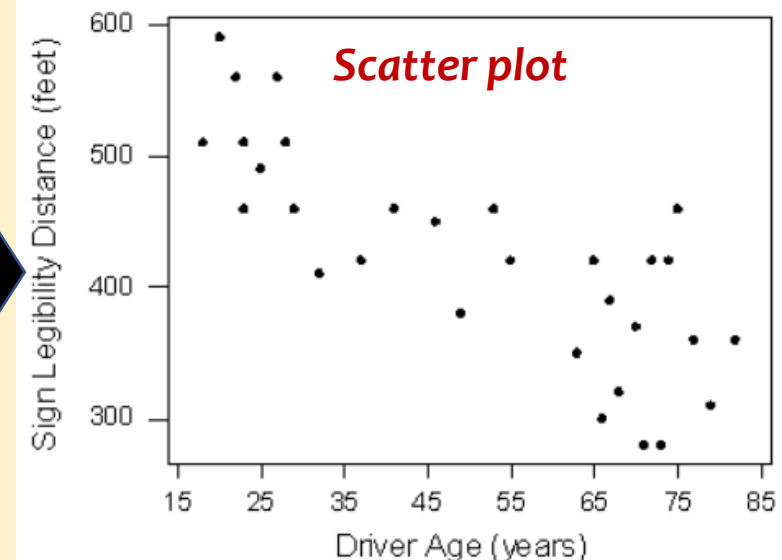
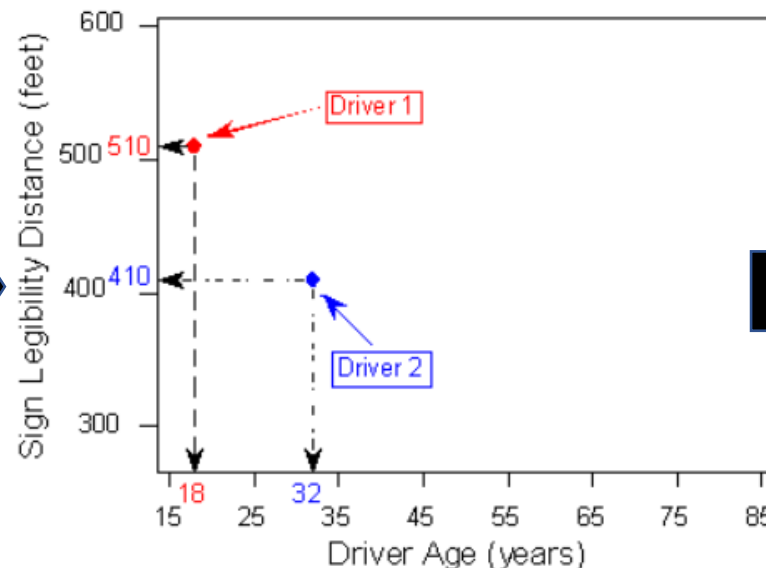
3

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

- In case $C \rightarrow Q$ we compared distributions of the quantitative response.
- In case $C \rightarrow C$ we compared distributions of the categorical response.

Case $Q \rightarrow Q$ is different in the sense that both variables (in particular the explanatory variable) are quantitative, and therefore, as you'll discover, this case will require a different kind of treatment and tools.

	Age (X)	Distance (Y)
Driver 1	18	510
Driver 2	32	410
Driver 3	55	420
Driver 4	23	510
.	.	.
.	.	.
.	.	.
Driver 30	82	360



Examining Relationships

- The appropriate and informative graphical display for examining the relationship between ***two quantitative variables*** is the **scatterplot**

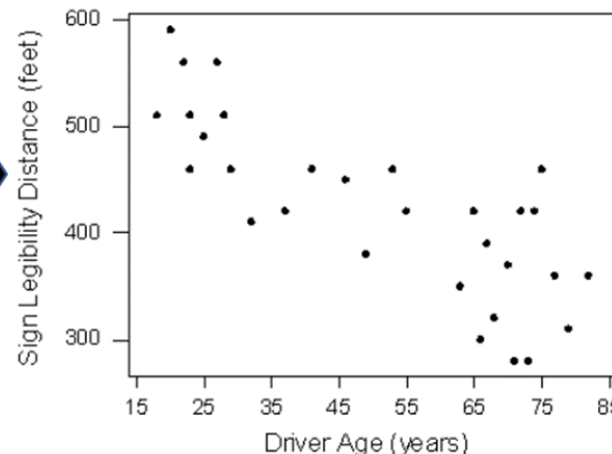
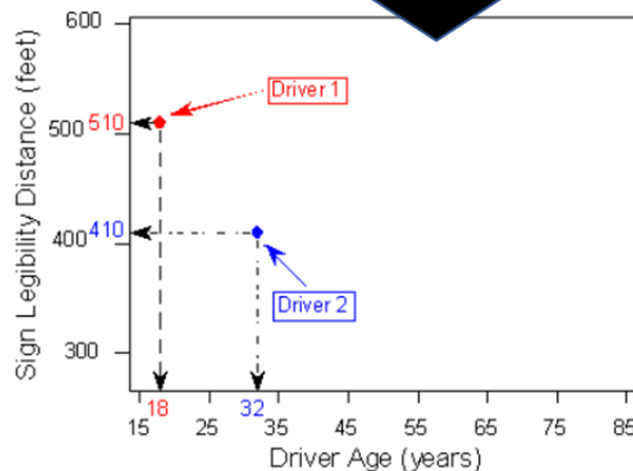
Explanatory

Response

How to construct a **scatterplot**?

- Each pair of values is plotted, so that the value of the **explanatory variable (X)** is plotted on the **horizontal axis**, and the value of the **response variable (Y)** is plotted on the **vertical axis**.
- In short, each individual (here, driver) appears on the **scatterplot** as a *single point* whose X-coordinate is the value of the explanatory variable for that individual, and whose Y-coordinate is the value of the response variable.

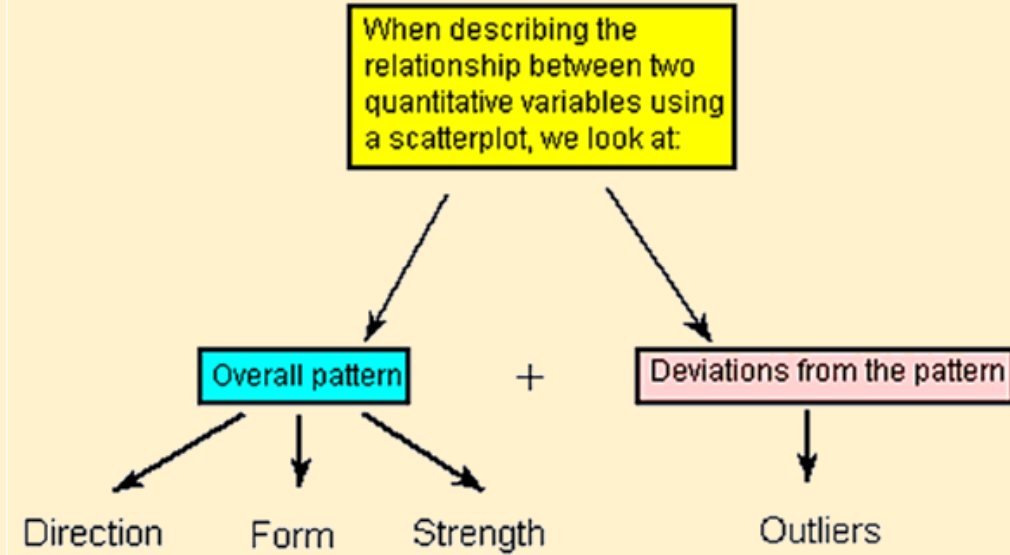
	Age (X)	Distance (Y)
Driver 1	18	510
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.	.	.
.	.	.
.	.	.
Driver 30	82	360



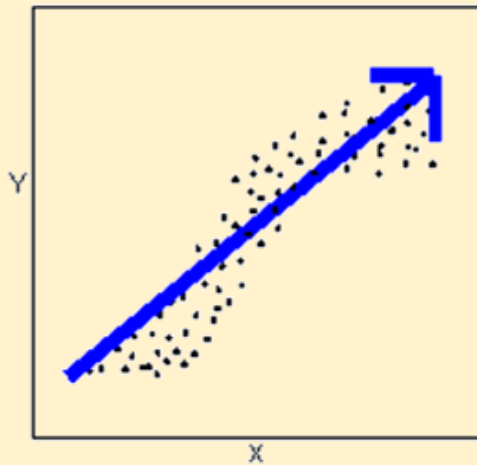
Reading Exercise – No 16 (Module | Statistical Reasoning 2 of 4)

3

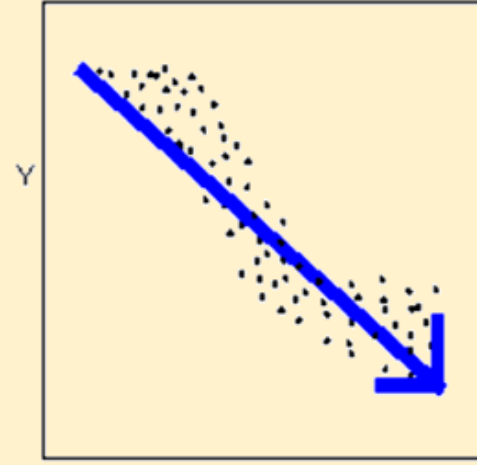
		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q



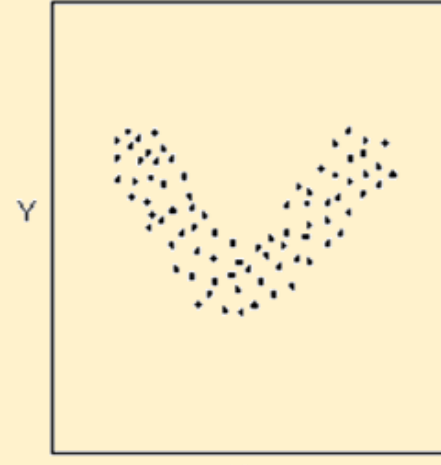
1. The direction of the relationship can be positive, negative, or neither:



Positive Relationship



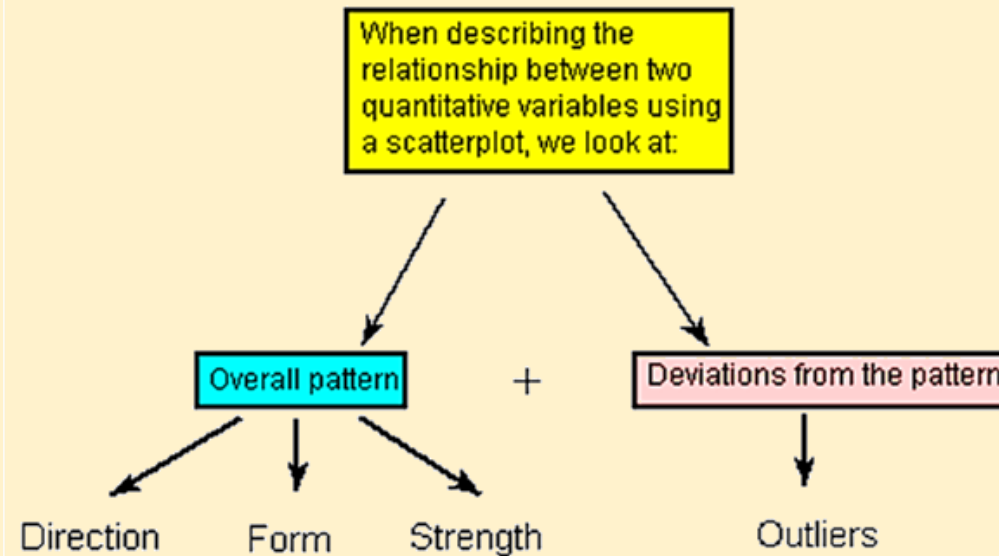
Negative Relationship



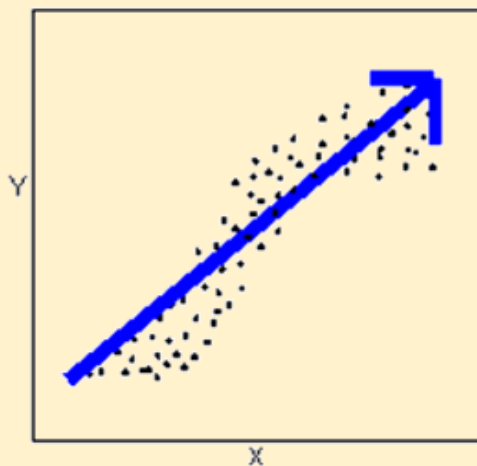
Neither Positive nor Negative Relationship

3

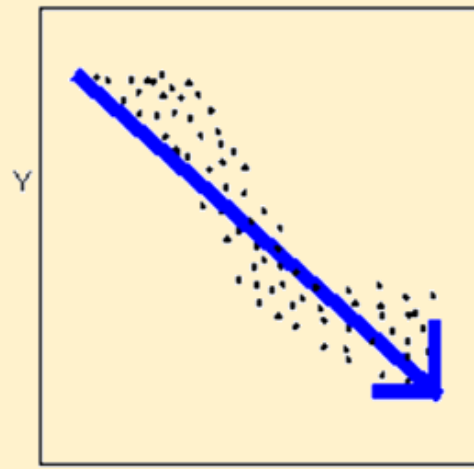
		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q



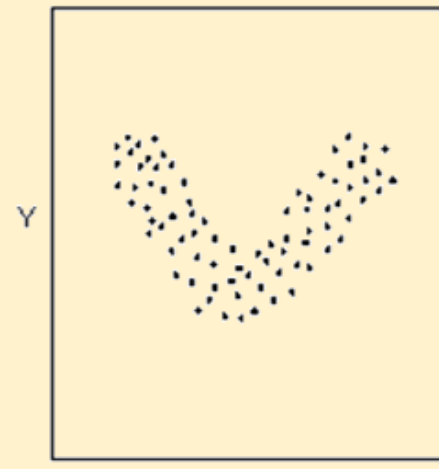
1. The direction of the relationship can be positive, negative, or neither:



Positive Relationship



Negative Relationship

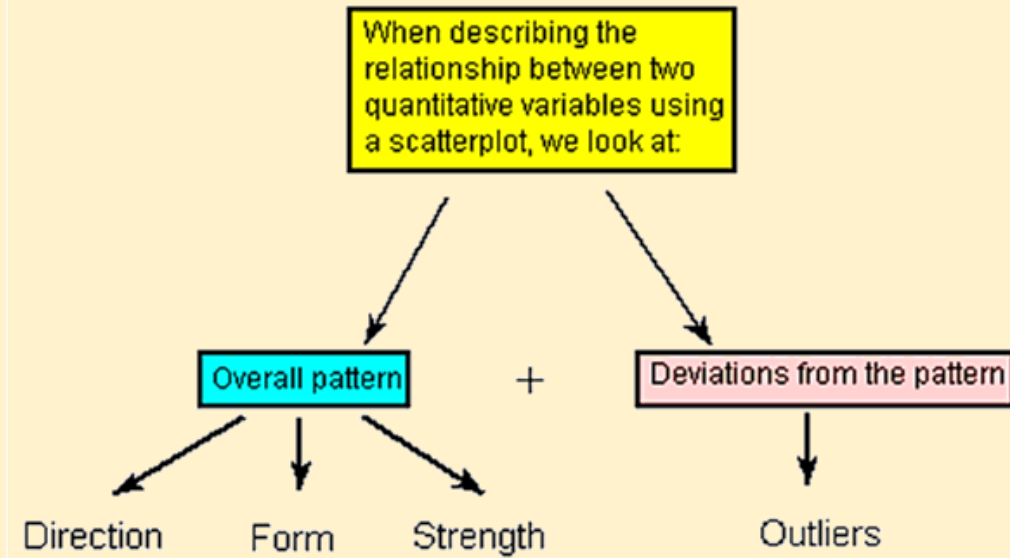


Neither Positive nor Negative Relationship

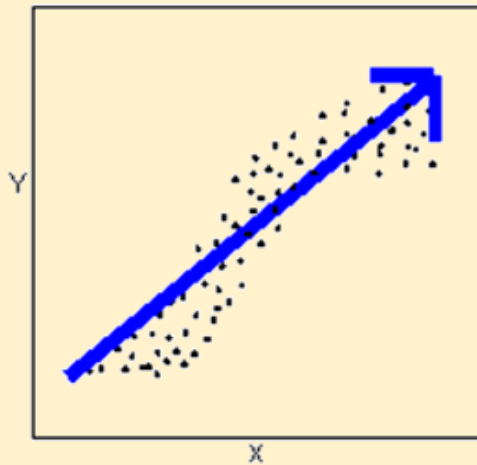
A **positive (or increasing) relationship** means that an increase in one of the variables is associated with an increase in the other.

3

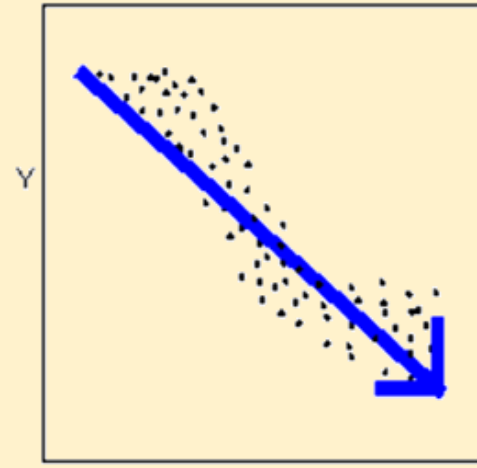
		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q



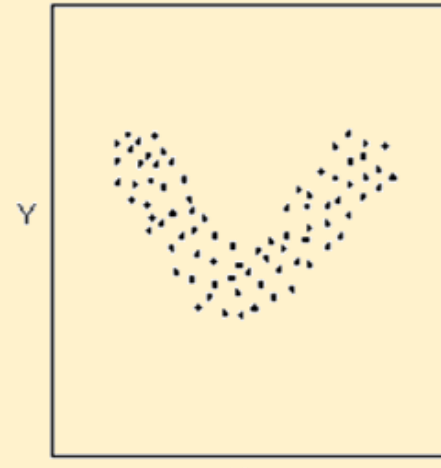
1. The direction of the relationship can be positive, negative, or neither:



Positive Relationship



Negative Relationship



Neither Positive nor Negative Relationship

A **negative (or decreasing relationship)** means that an increase in one of the variables is associated with an decrease in the other.

Reading Exercise – No 16 (Module | Statistical Reasoning 2 of 4)

3

Response

Categorical

Quantitative

Categorical

Quantitative

C → C

C → Q

Q → C

Q → Q

When describing the relationship between two quantitative variables using a scatterplot, we look at:

Overall pattern

+

Deviations from the pattern

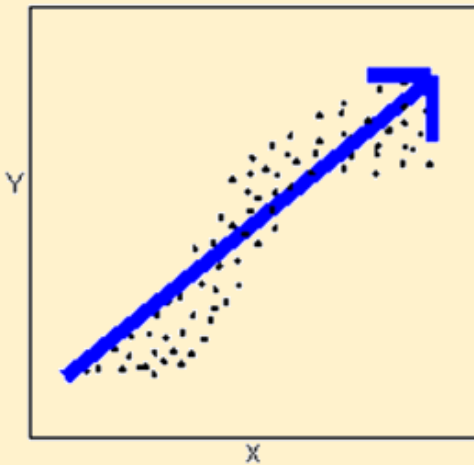
Direction

Form

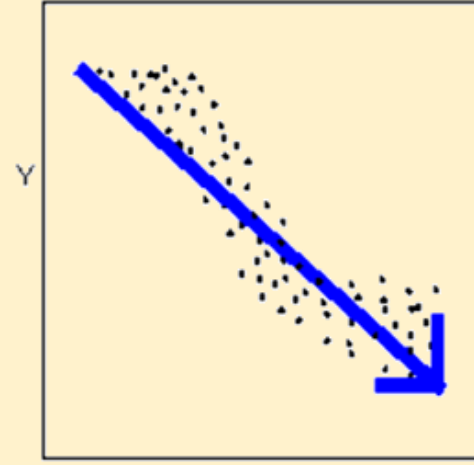
Strength

Outliers

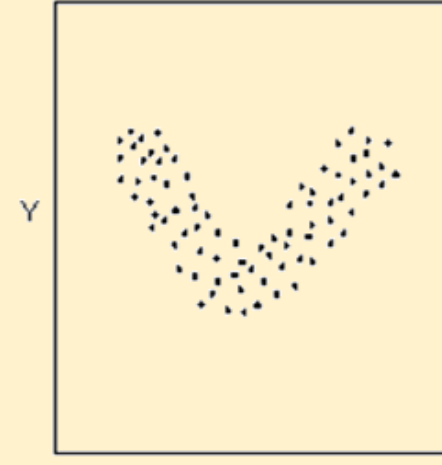
1. The direction of the relationship can be positive, negative, or neither:



Positive Relationship



Negative Relationship

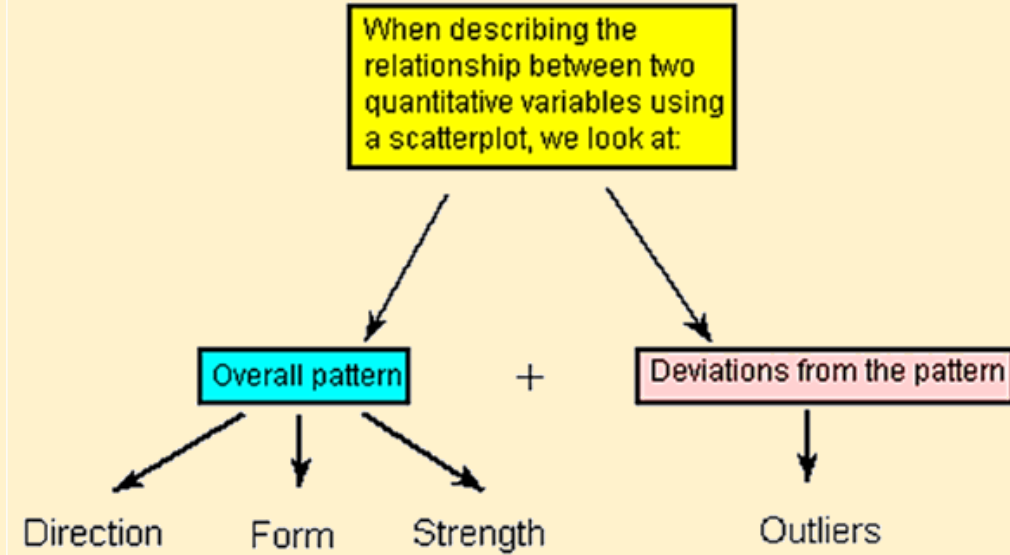


Neither Positive nor Negative Relationship

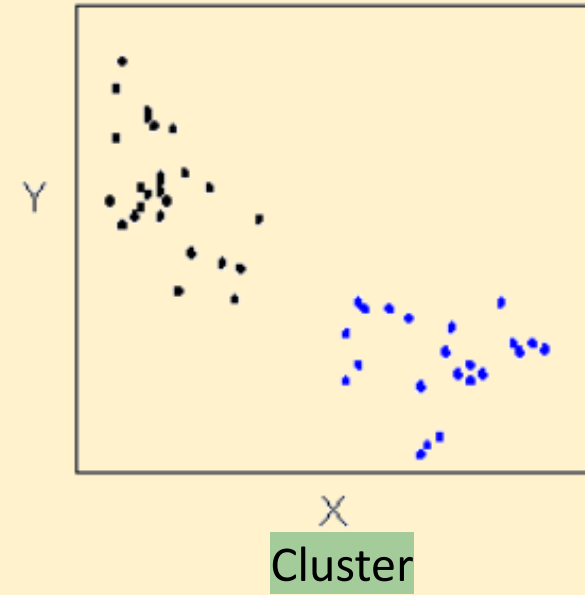
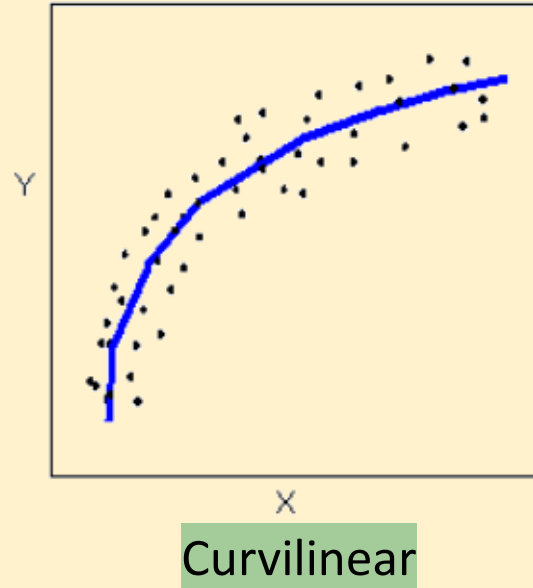
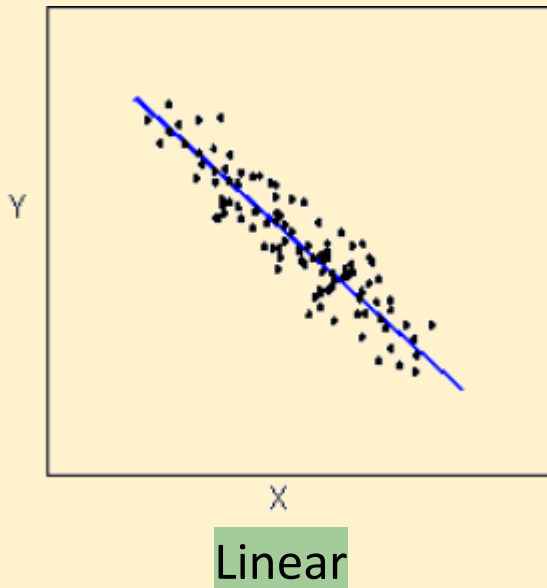
Not all relationships can be classified as either positive or negative.

3

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q



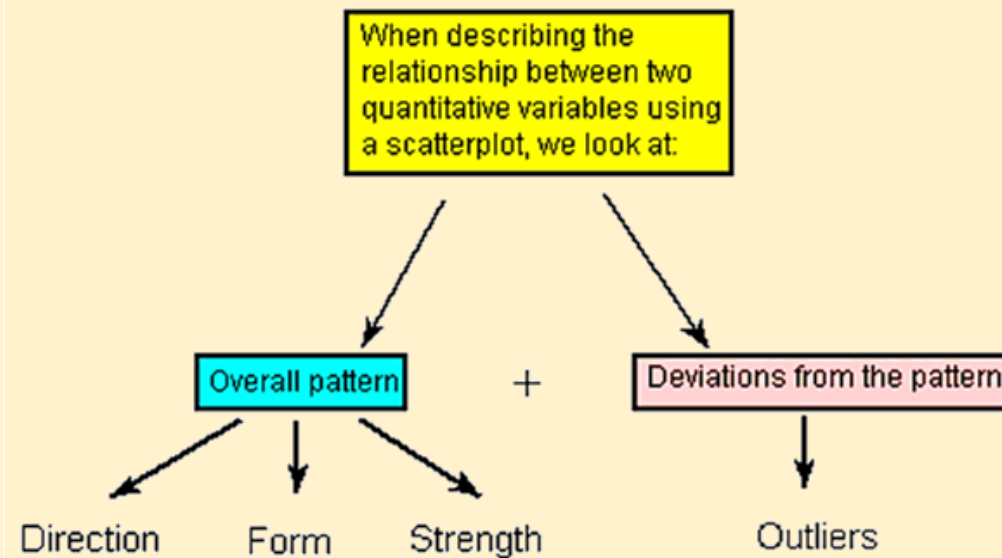
2. The form of the relationship is its **general shape** – Linear, Curvilinear, Cluster



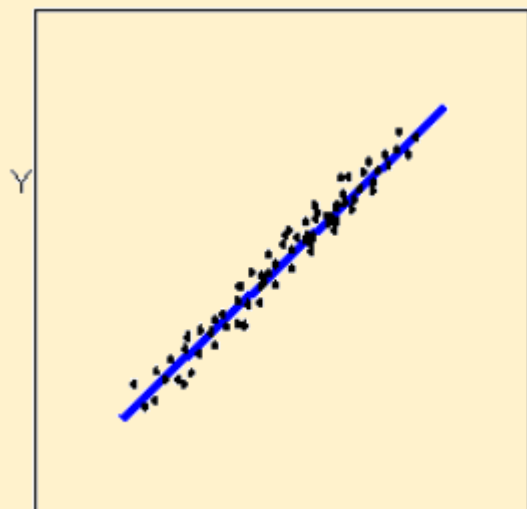
Reading Exercise – No 16 (Module | Statistical Reasoning 2 of 4)

3

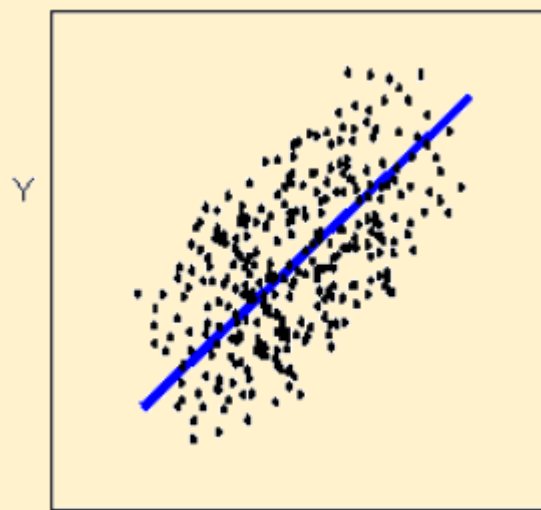
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		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q



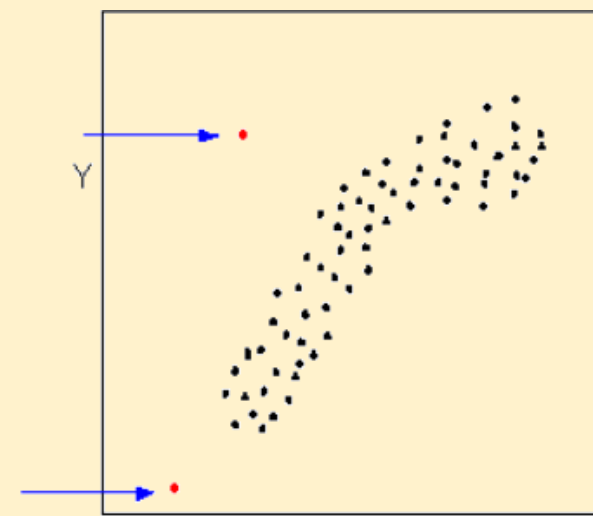
3. The strength of the relationship is determined by how closely the data follow the form of the relationship.



strong relationship



weaker relationship



outliers

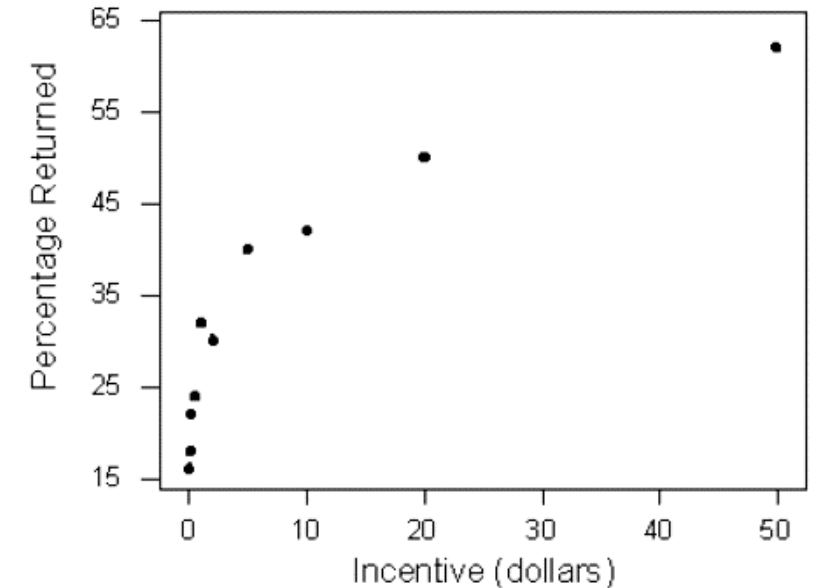
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – The Role-Type Classification – 3 (Scatterplot)

A study examined how the percentage of participants who completed a survey is related to the monetary incentive that researchers promised to participants. Consider the relationship between these two quantitative variables ($Q \rightarrow Q$), displayed in the scatterplot:

Question No 01 – The **direction** of this relationship is {positive, negative, neither positive nor negative}.



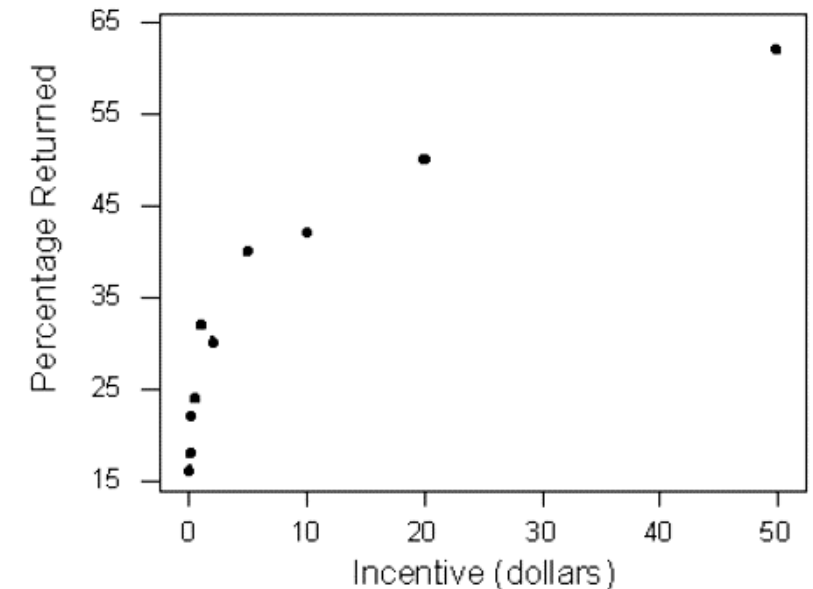
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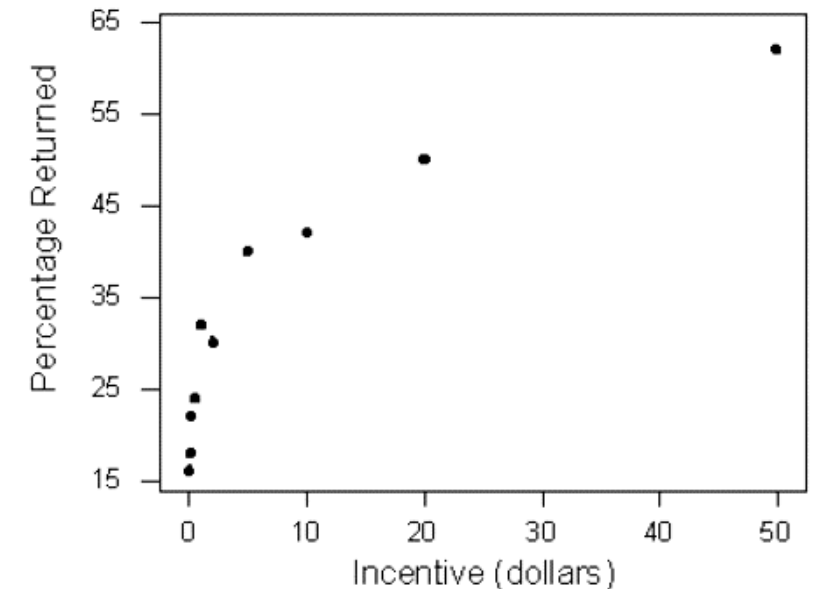
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A study examined how the percentage of participants who completed a survey is related to the monetary incentive that researchers promised to participants. Consider the relationship between these two quantitative variables ($Q \rightarrow Q$), displayed in the scatterplot:

Question No 02 – In the context of this example, this means that when researchers promised higher payments, the percentage of participants who **completed** the survey {increased, remained the same, decreased}



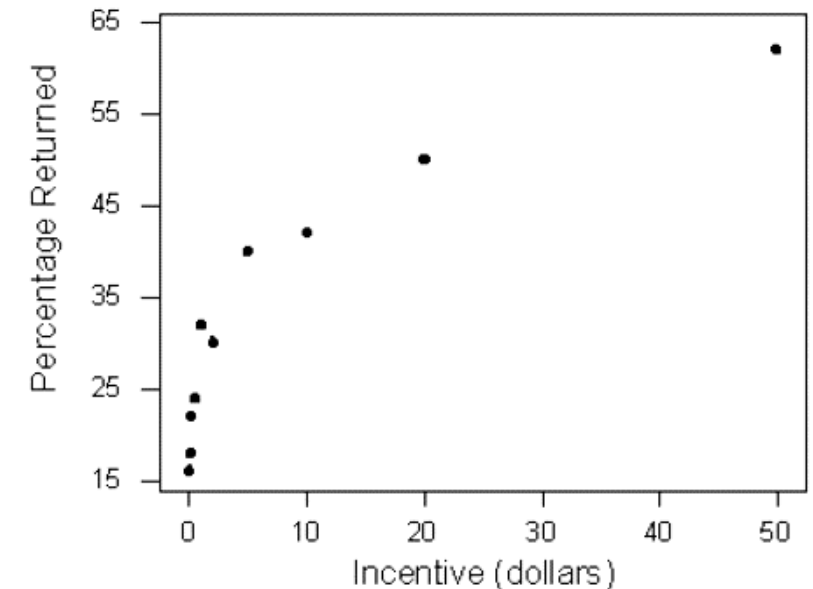
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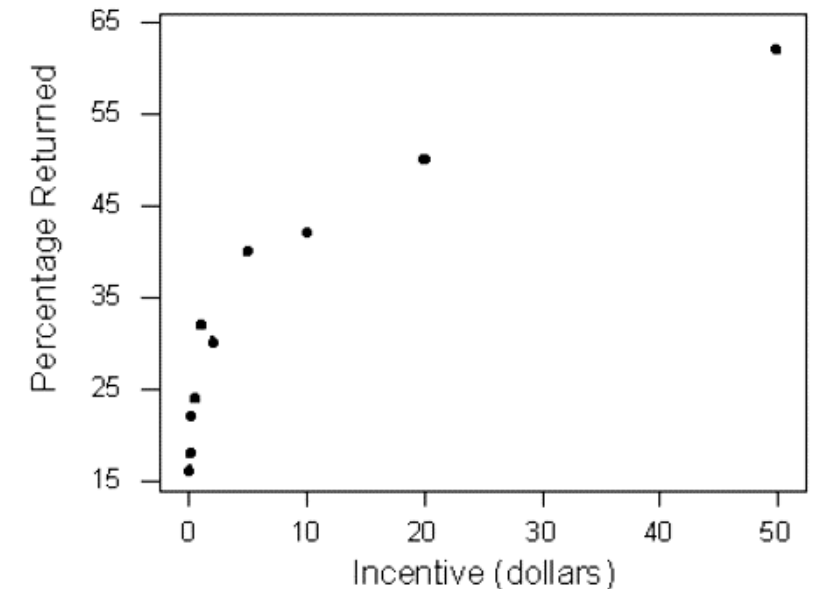
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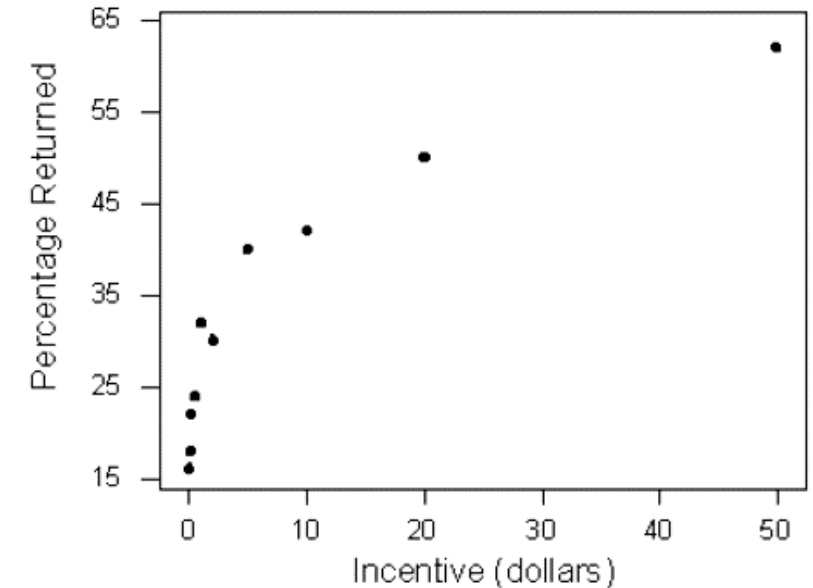
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – The Role-Type Classification – 3 (Scatterplot)

A study examined how the percentage of participants who completed a survey is related to the monetary incentive that researchers promised to participants. Consider the relationship between these two quantitative variables ($Q \rightarrow Q$), displayed in the scatterplot:

Question No 03 – The **form** of the relationship is {linear, curvilinear, neither linear nor curvilinear}.



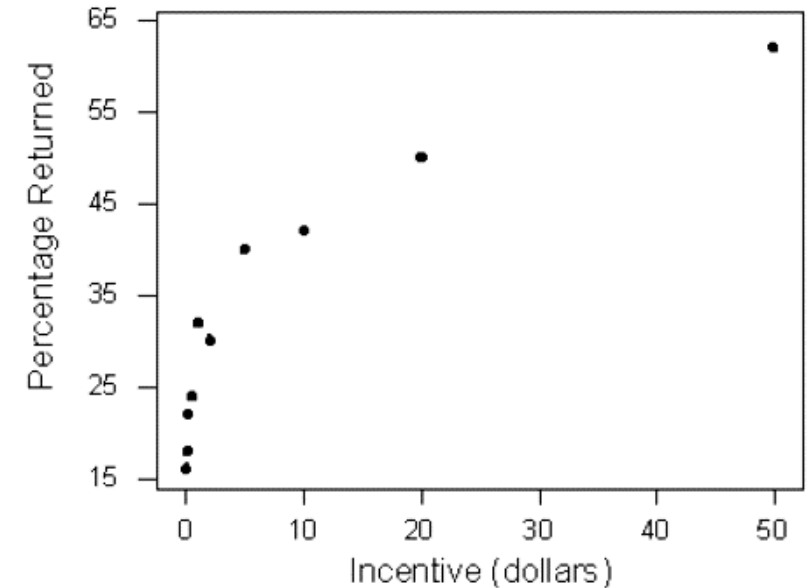
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – The Role-Type Classification – 3 (Scatterplot)

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Question No 03 – The **form** of the relationship is {linear, **curvilinear**, neither linear nor curvilinear}.



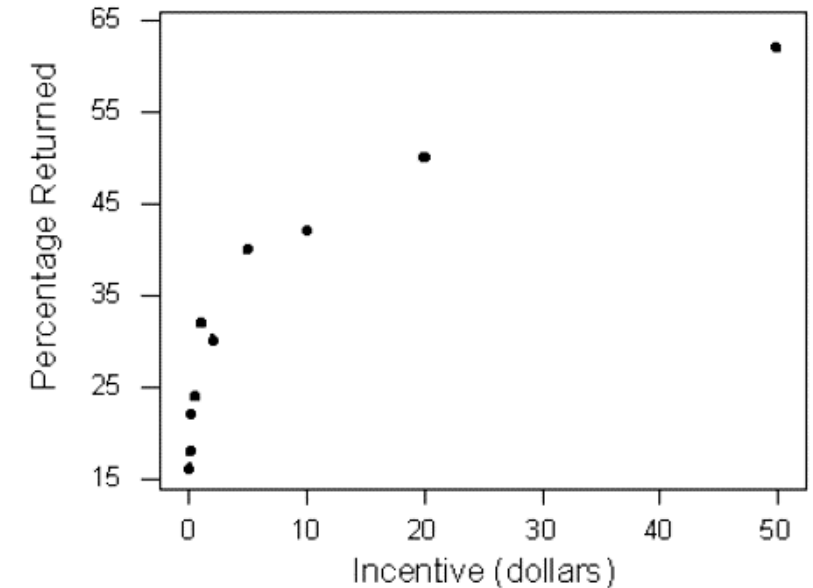
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – The Role-Type Classification – 3 (Scatterplot)

A study examined how the percentage of participants who completed a survey is related to the monetary incentive that researchers promised to participants. Consider the relationship between these two quantitative variables ($Q \rightarrow Q$), displayed in the scatterplot:

Question No 04 – Based on the form of the relationship as illustrated in the scatterplot, the **relationship** is quite {strong, weak}.



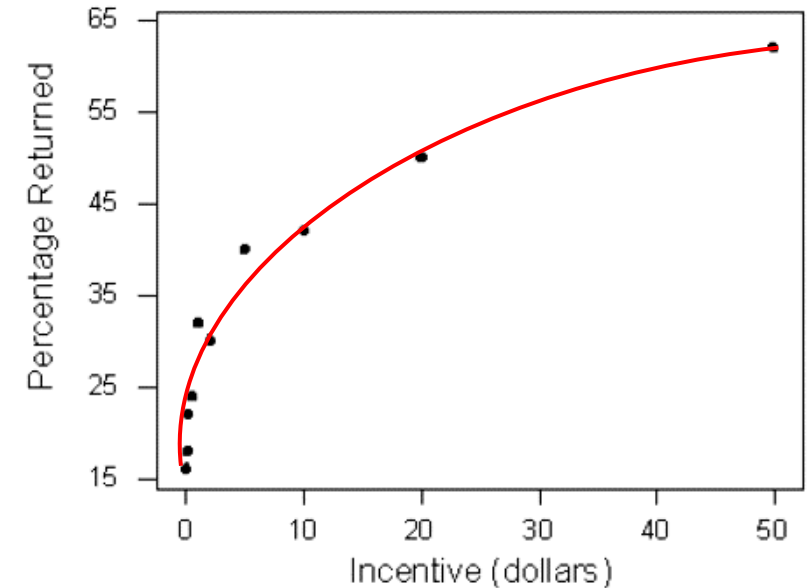
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

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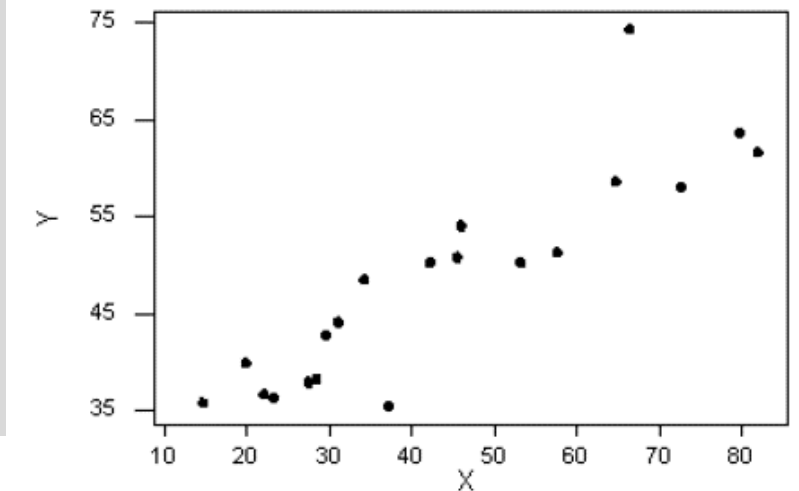
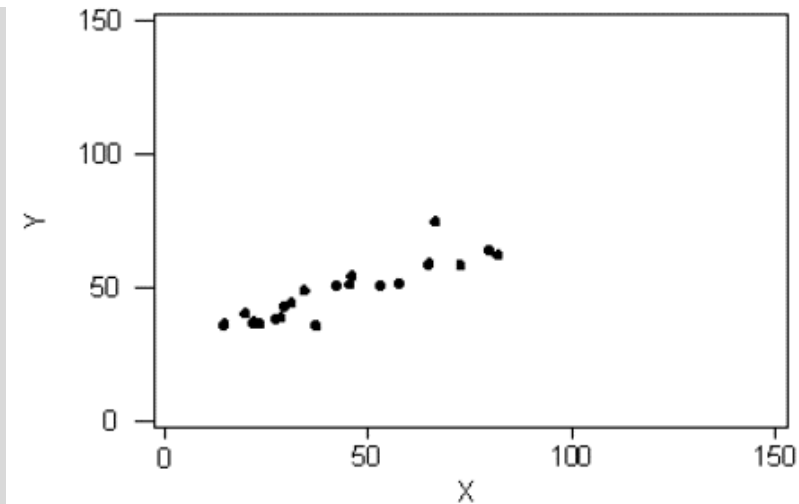
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss – Linear Relationship

Question No 05 – Which of the two scatterplots do you think displays a **stronger** linear relationship?

- A. The top scatterplot.
- B. The bottom scatterplot.
- C. Both relationships are equally strong.



Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

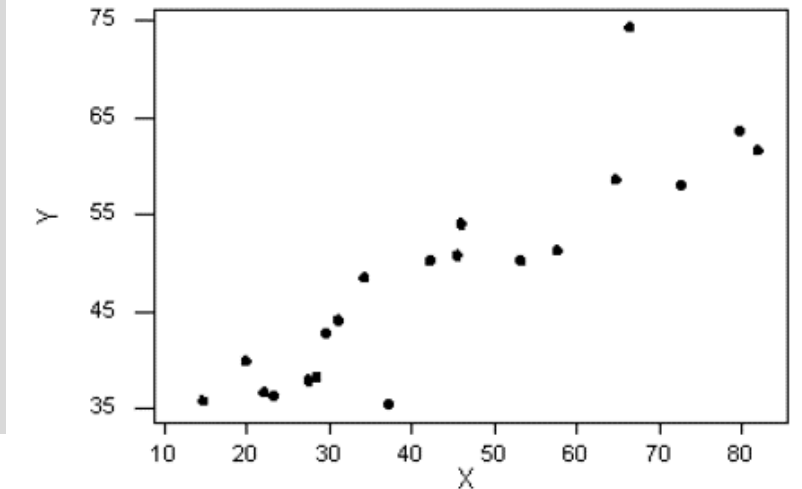
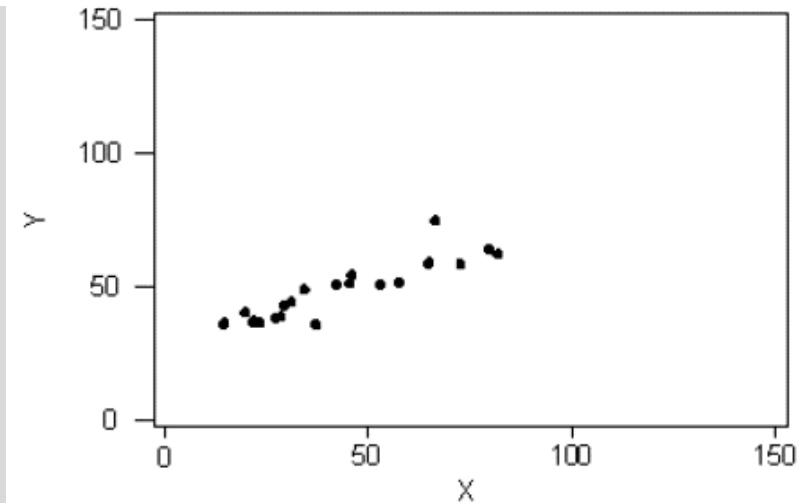
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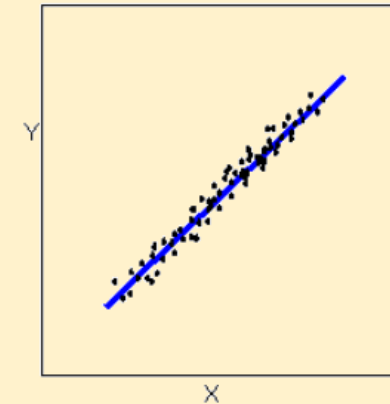
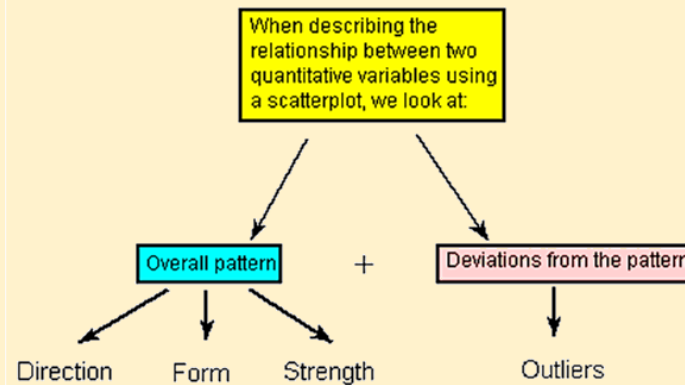
Answer C

Both scatterplots display the same data, but they look different, because different scales were used on the axes.

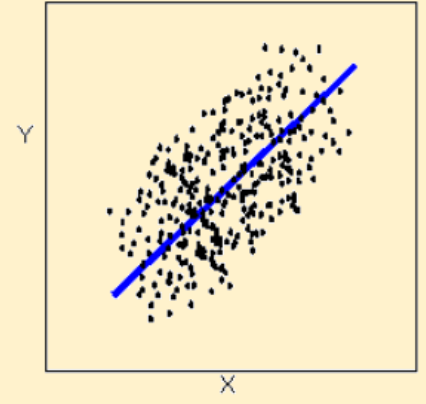


3

		Response	
		Categorical	Quantitative
Explanatory	Categorical	$C \rightarrow C$	$C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$



strong relationship



weaker relationship

❑ **The Correlation Coefficient – r** The numerical measure that assesses the strength of a linear relationship is called **the correlation coefficient** and is denoted by r .

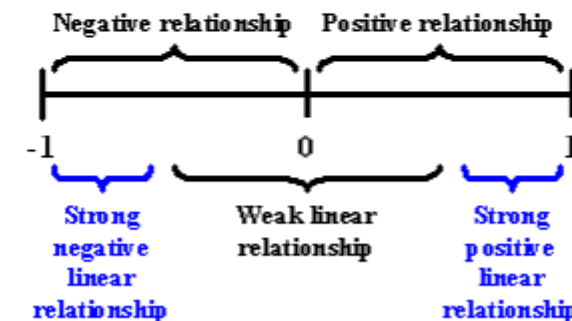
❑ **The Correlation Coefficient – r** -1 to +1

❑ Close to -1/+1 – strong

❑ Close to 0 – weak

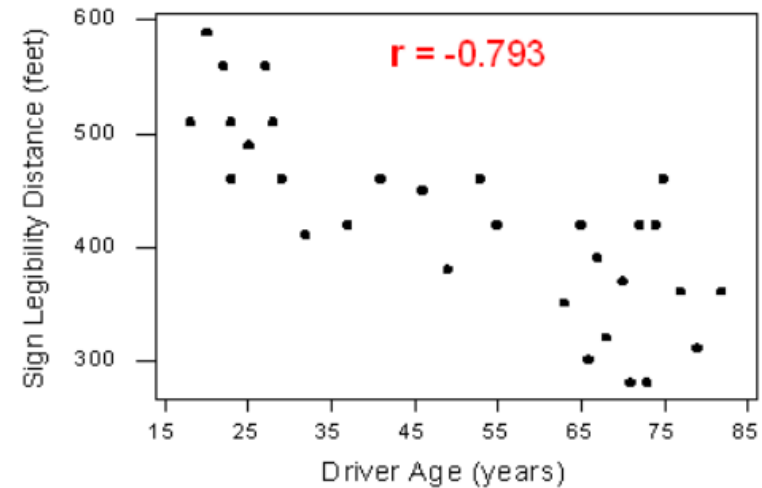
❑ Close to -0,5/+0.5 – moderately strong

Values that are close to -1 or close to 1 indicate a strong linear relationship, either negative or positive.



Example (F) Highway Sign Visibility

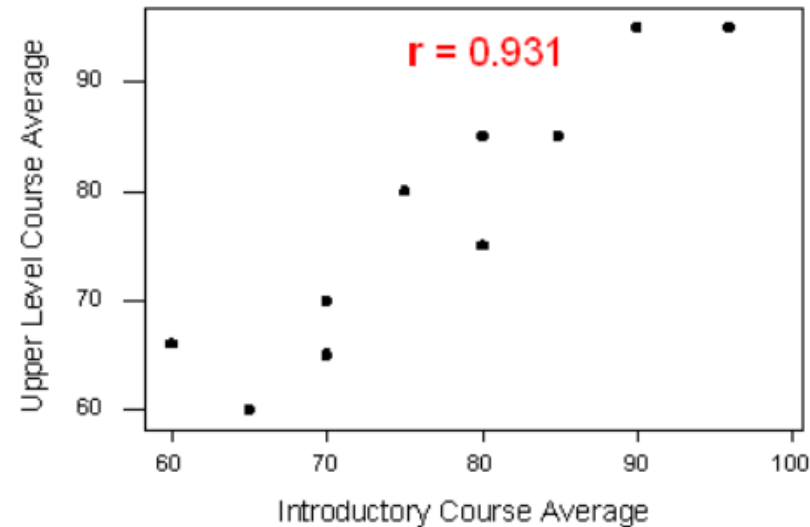
Earlier, we used the scatterplot below to find a *negative linear* relationship between the age of a driver and the maximum distance at which a highway sign was legible. What about the strength of the relationship? It turns out that the correlation between the two variables is $r = -0.793$.



- Since $r < 0$, it confirms that the direction of the relationship is **negative**
- Since r is relatively close to -1 , it suggests that the relationship is **moderately strong**
 - In the context, the negative correlation confirms that the maximum distance at which a sign is legible generally decreases with age.
 - Since the value of r indicates that the linear relationship is moderately strong, we can expect the maximum distance to vary somewhat, even among drivers of the same age.

Example (G) Statistics Courses

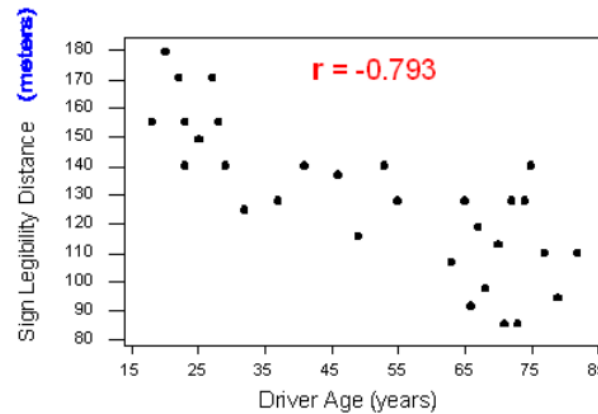
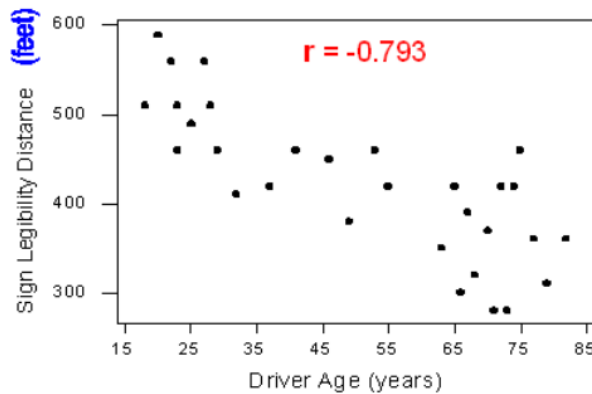
A statistics department is interested in tracking the progress of its students from entry until graduation. As part of the study, the department tabulates the performance of 10 students in an introductory course and in an upper-level course required for graduation. What is the relationship between the students' course averages in the two courses? Here is the scatterplot for the data:



- The scatterplot suggests a relationship that is **positive** in direction, **linear** in form, and seems quite **strong**
- The value of the correlation that we find between the two variables is $r = 0.931$, which is very close to 1, and thus confirms that indeed the linear relationship is **very strong**.

Examining Relationships

- There are several important properties of the **correlation coefficient r** as a **numerical measure of the strength of a linear relationship**:
 1. The correlation does not change when the units of measurement of either one of the variable change. In other words, if we change the units of measurement of the explanatory variable and/or the response variable, the change has *no effect on the correlation (r)*.

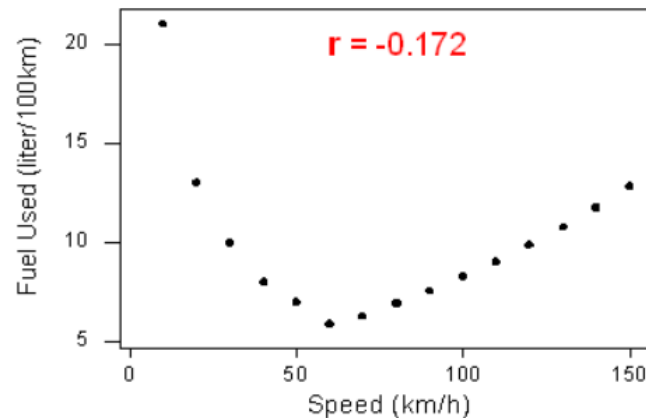


Notice that the Y-values have changed, but the correlations are the same.

Examining Relationships

- There are several important properties of the **correlation coefficient r** as a **numerical measure of the strength of a linear relationship**:
 2. The correlation measures only the **strength** of a linear relationship between two variables. It **ignores** any other type of relationship, no matter how strong it is.

E.g., consider the relationship between the average fuel usage of driving a fixed distance in a car, and the speed at which the car drives:

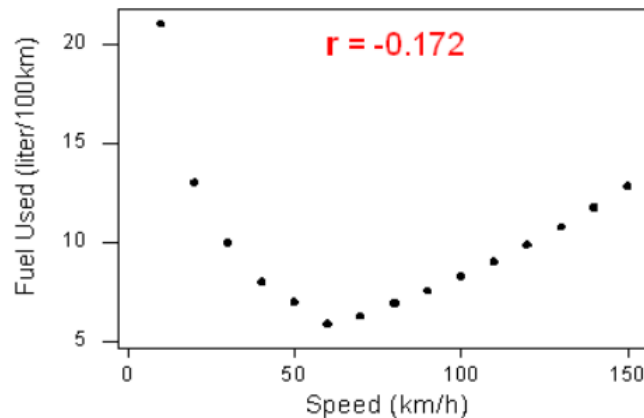


The data describe a fairly **simple curvilinear relationship**: the amount of fuel consumed decreases rapidly to a minimum for a car driving 60 kilometers per hour, and then increases gradually for speeds exceeding 60 kilometers per hour. The relationship is **very strong**, as the observation seems to perfectly fit the curve.

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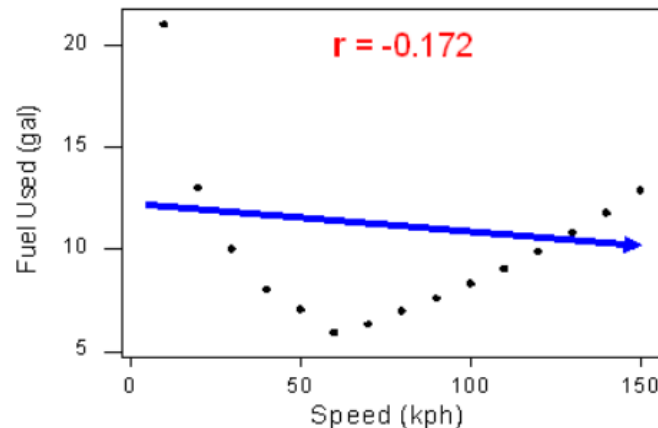


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Although the relationship is strong, the correlation $r = -0.172$ indicates a **weak linear** relationship ...

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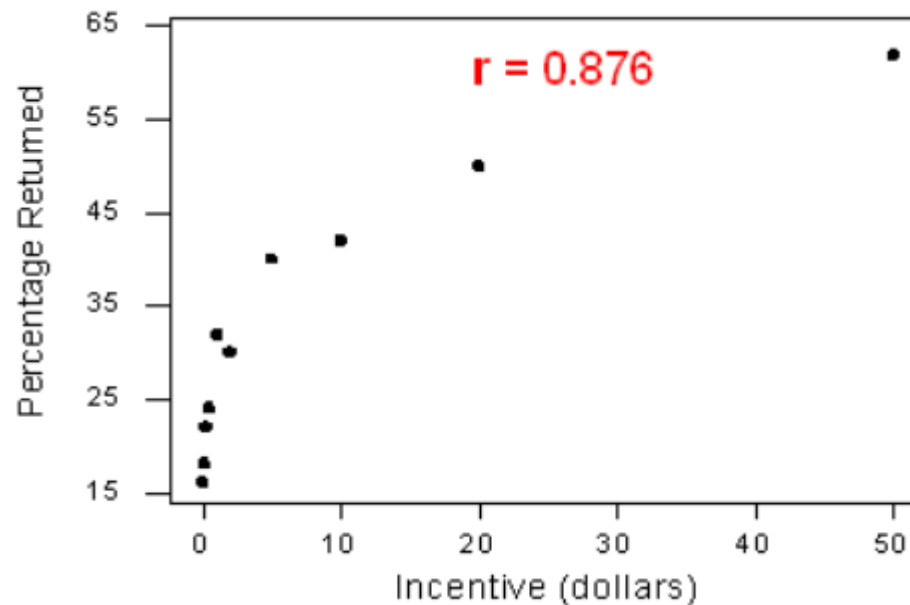
This makes sense considering that the data fails to adhere closely to a linear form.

Although the relationship is strong, the correlation $r = -0.172$ indicates a **weak linear** relationship ...

Examining Relationships

- There are several important properties of the **correlation coefficient r** as a **numerical measure of the strength of a linear relationship**:
 - The correlation by itself is **not sufficient** to determine whether the relationship is linear.

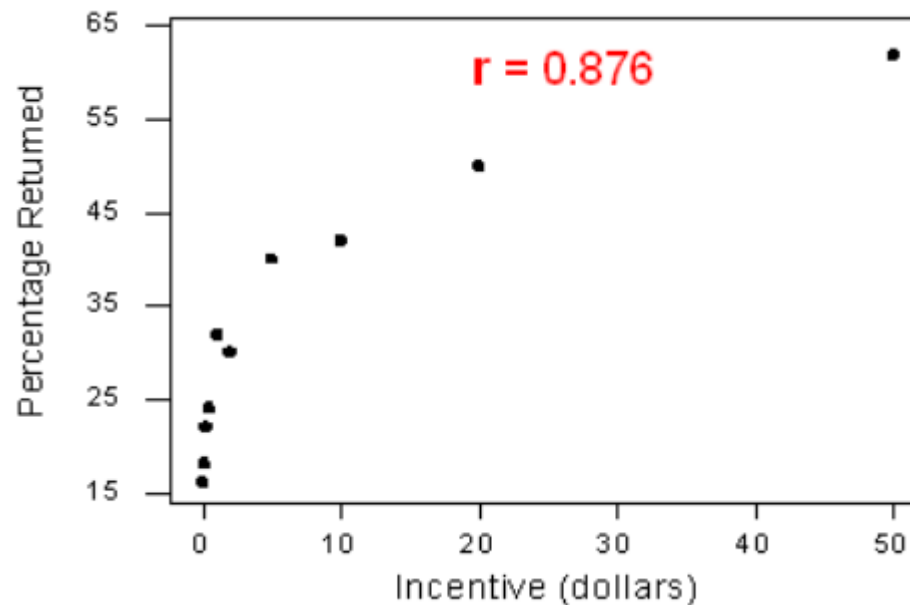
E.g., consider the study that examined the effect of monetary incentives on the return rate of questionnaires. On the right is the scatterplot relating the percentage of participants who completed a survey to the monetary incentive that researchers promised to participants, in which we find a strong curvilinear relationship:



Examining Relationships

- There are several important properties of the **correlation coefficient r** as a **numerical measure of the strength of a linear relationship**:
 - The correlation by itself is ***not sufficient*** to determine whether the relationship is linear.

E.g., consider the study that examined the effect of monetary incentives on the return rate of questionnaires. On the right is the scatterplot relating the percentage of participants who completed a survey to the monetary incentive that researchers promised to participants, in which we find a ***strong curvilinear relationship***:



The relationship is curvilinear, yet the correlation $r = 0.876$ is quite close to 1.

Therefore, the correlation alone does not indicate whether a relationship is linear.

Always look at the data!

Examining Relationships

- There are several important properties of the correlation coefficient r as a numerical measure of the strength of a linear relationship:
 4. The correlation is heavily influenced by outliers. The way in which the outlier influences the correlation depends on whether the outlier is consistent with the pattern of the linear relationship.

Examining Relationships

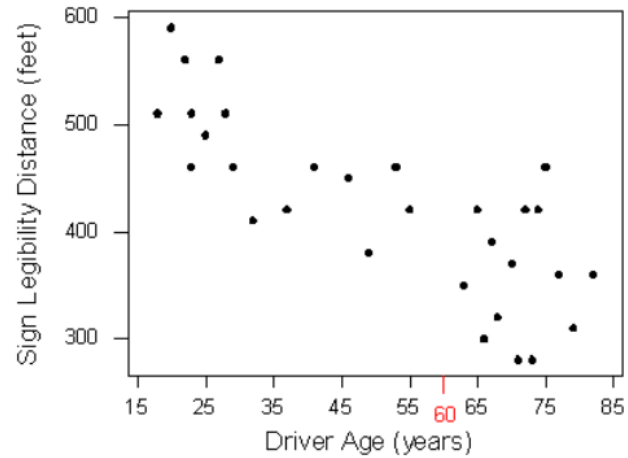
- So far we've used the scatterplot to describe the relationship between two quantitative variables, and in the special case of a linear relationship, we have supplemented the scatterplot with the correlation (r).
- The correlation, however, doesn't fully characterize the linear relationship between two quantitative variables – *it only measures the strength and direction*.

Examining Relationships

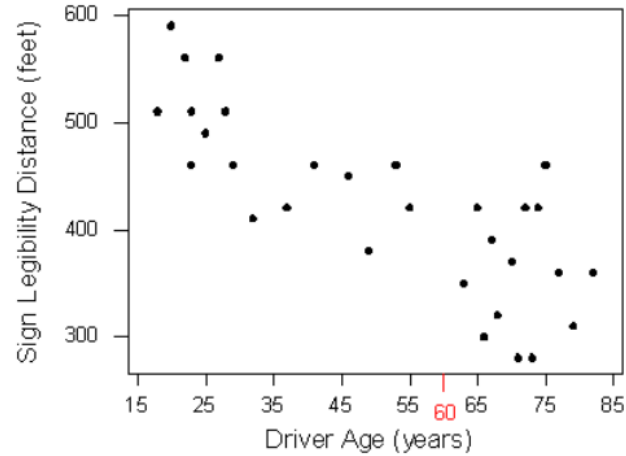
- We often want to describe more ***precisely*** how one variable changes with the other, or ***predict*** the value of the **response variable** for a given value of the **explanatory variable**.
- In order to be able to do that, we need to summarize the linear relationship ***with a line that best fits the linear pattern of the data.***

Examining Relationships

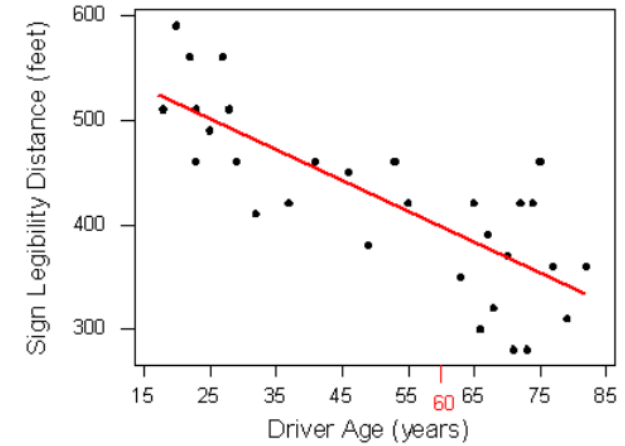
Here again is the scatterplot of driver ages and maximum distances at which a sign is legible. The age for which an agency wishes to predict the legibility distance, 60, is marked in red. How can we make this prediction?



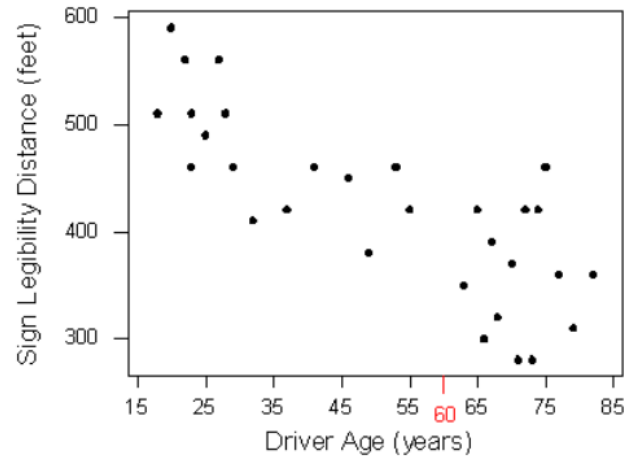
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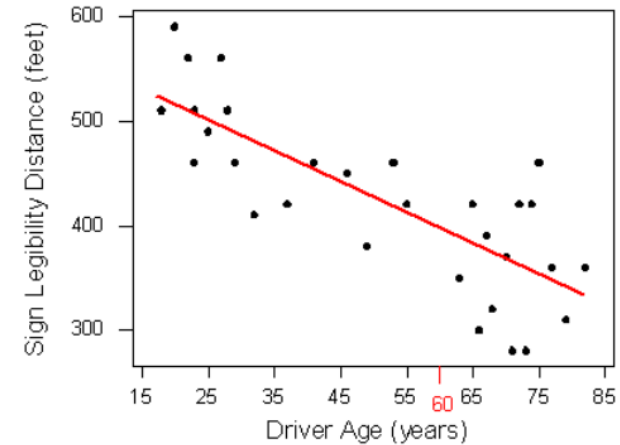
- It would be useful if we could find a line that represents the general pattern of the data ...



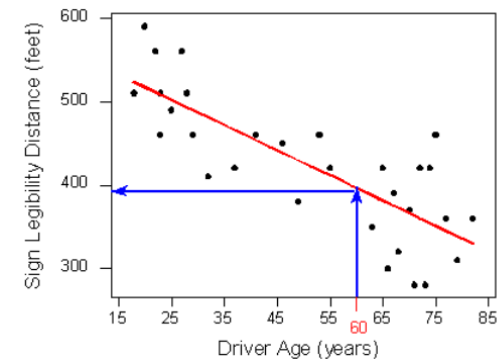
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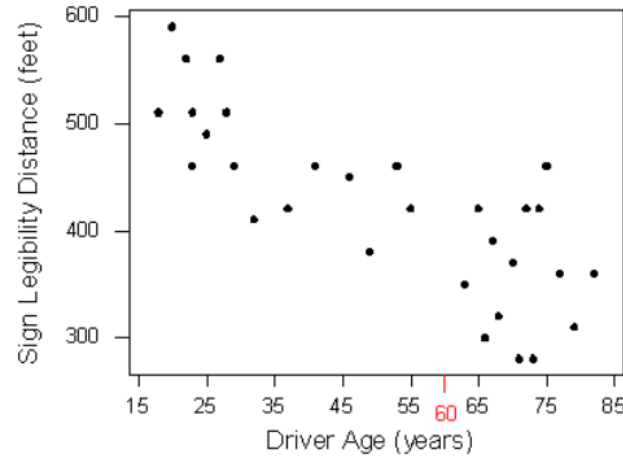


...because then, we would simply use this line to find the distance that corresponds to an age of 60 like this:



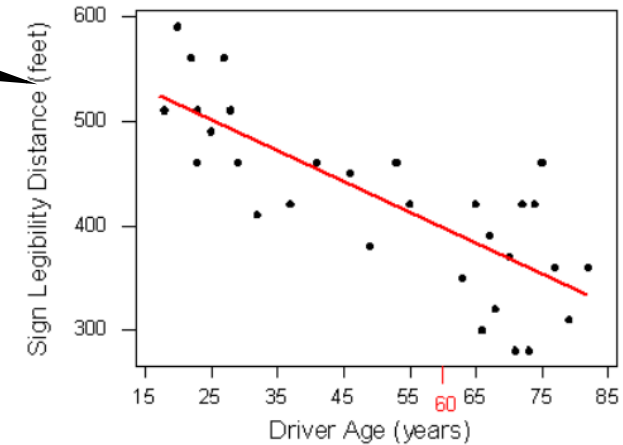
And predict that 60-year-old drivers could see the sign from a distance of just under 400 feet.

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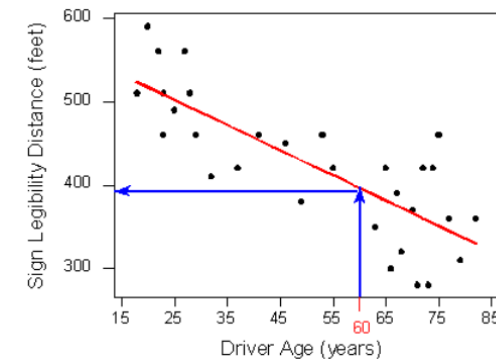


How and why did we pick this particular line? What line exactly did we choose?

- It would be useful if we could find a line that represents the general pattern of the data ...



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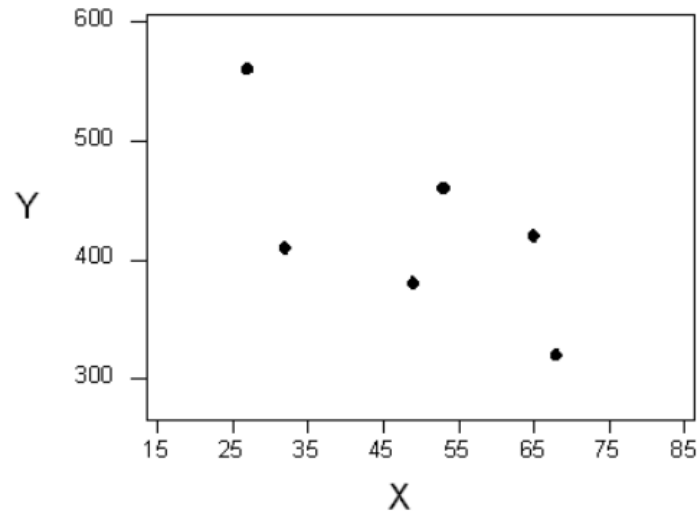
And predict that 60-year-old drivers could see the sign from a distance of just under 400 feet.

Examining Relationships

- **Linear regression** is the technique of finding the line that best fits the pattern of the linear relationship (or, in other words, the line that best describes how the response variable linearly depends on the explanatory variable).

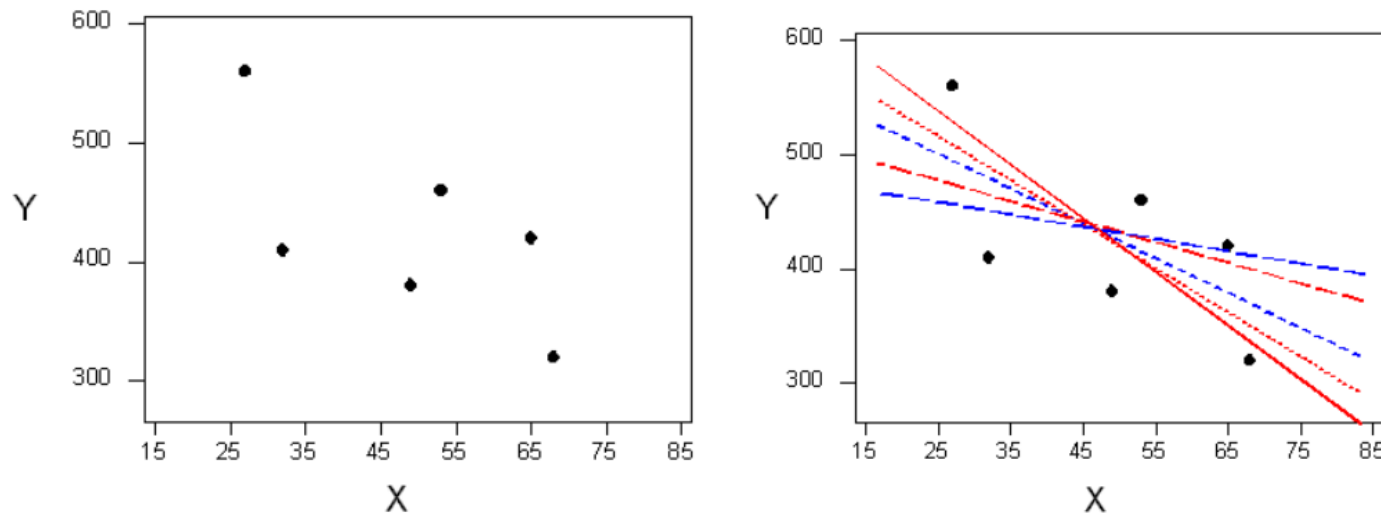
Examining Relationships

- Consider the following very simplified version of the age-distance example:



Examining Relationships

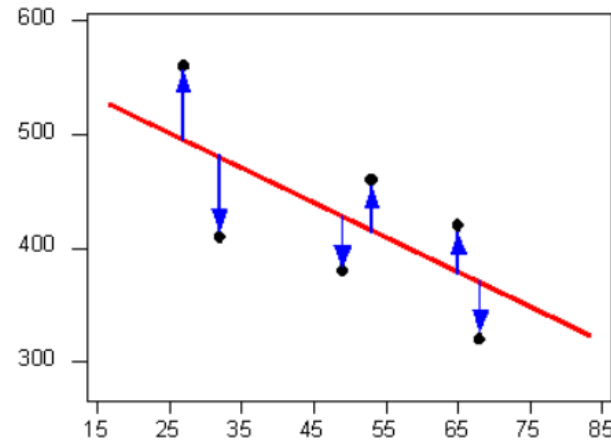
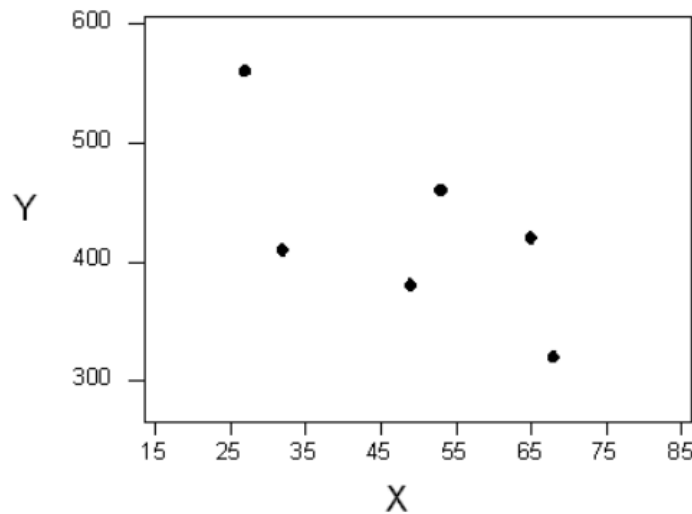
- Consider the following very simplified version of the age-distance example:



- There are many lines that look like they would be good candidates to be the line that best fits the data.
- However, it is doubtful that everyone would select the same line in the plot.
- Thus, we need to agree on what we mean by “best fits the data”
- In other words, we need a criterion by which we would select the line.

Examining Relationships

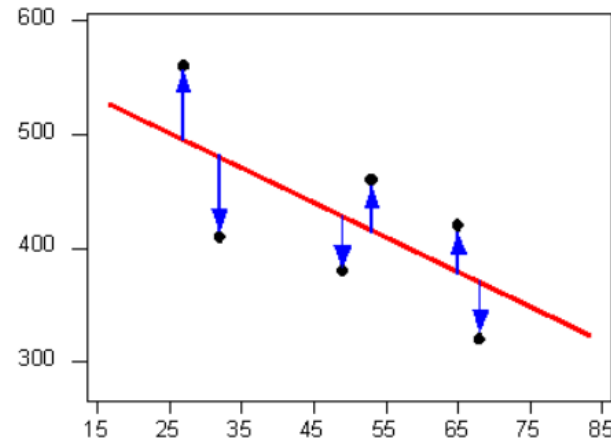
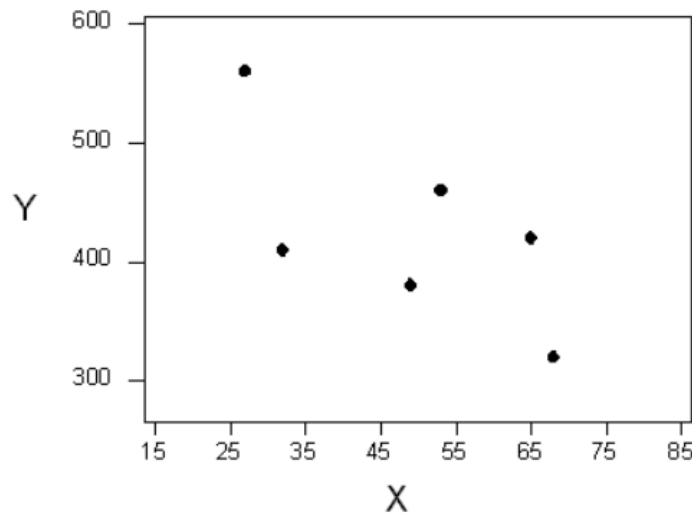
- Consider the following very simplified version of the age-distance example:



- We want the line we choose to be close to the data points.
- In other words, whatever criterion we choose, it had better somehow take into account the **vertical deviations** of the data points from the line, which are marked with the **blue arrows** in the plot on the left.

Examining Relationships

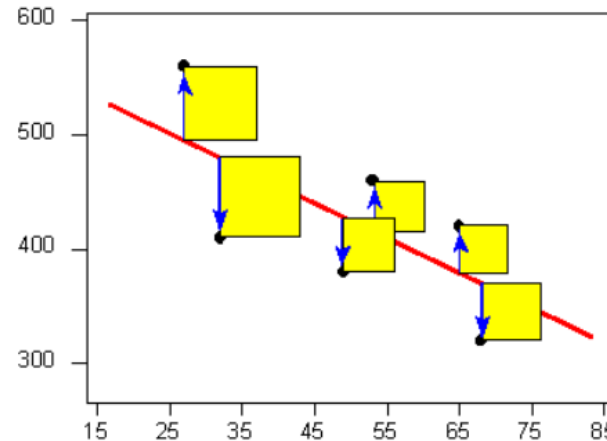
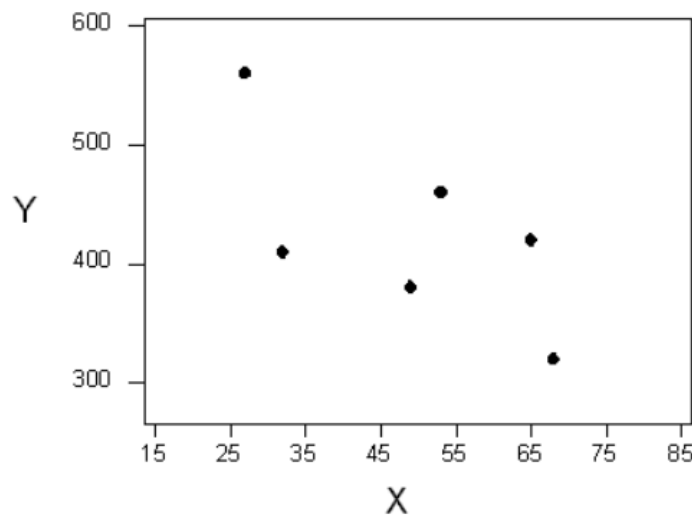
- Consider the following very simplified version of the age-distance example:



- The most commonly used criterion is called the **least squares criterion**:
 - Among all the lines that look good on your data, choose the one that has the smallest sum of squared vertical deviations.*

Examining Relationships

- Consider the following very simplified version of the age-distance example:



- The most commonly used criterion is called the **least squares criterion**:
 - Among all the lines that look good on your data, choose the one that has the smallest sum of squared vertical deviations.*
- Visually, each squared deviation is represented by the area of one of the squares in the plot on the left. Therefore, we are looking for the line that will have the smallest total yellow area.

Examining Relationships

- ***The algebra of a line***

- A line is described by a set of points (X,Y) that obey a particular relationship between X and Y. That relationship is called the **equation of the line**, which is expressed in the following form:

$$Y = a + bX$$

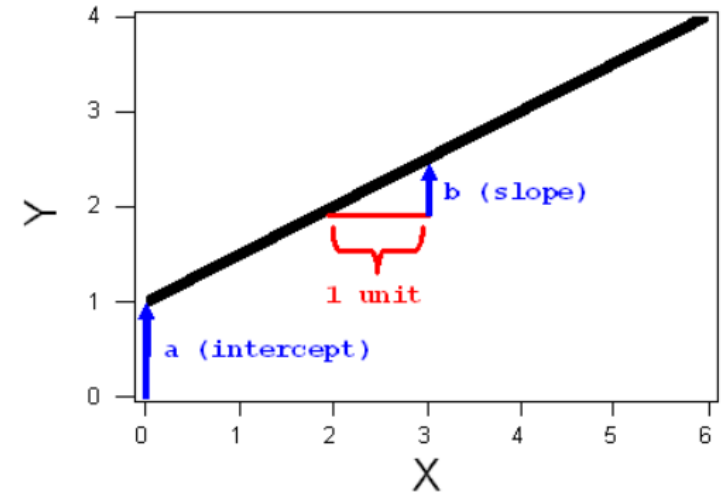
- In this equation **a** and **b** are constants that can be either **negative** or **positive**.
- The reason to write the line in this form is that the constants **a** and **b** tells us what the line looks like, as follows:
 1. The **intercept (a)** is the value that Y takes when $X = 0$
 2. The **slope (b)** is the change in Y for every increase of 1 unit in X.

Examining Relationships

$$Y = a + bX$$

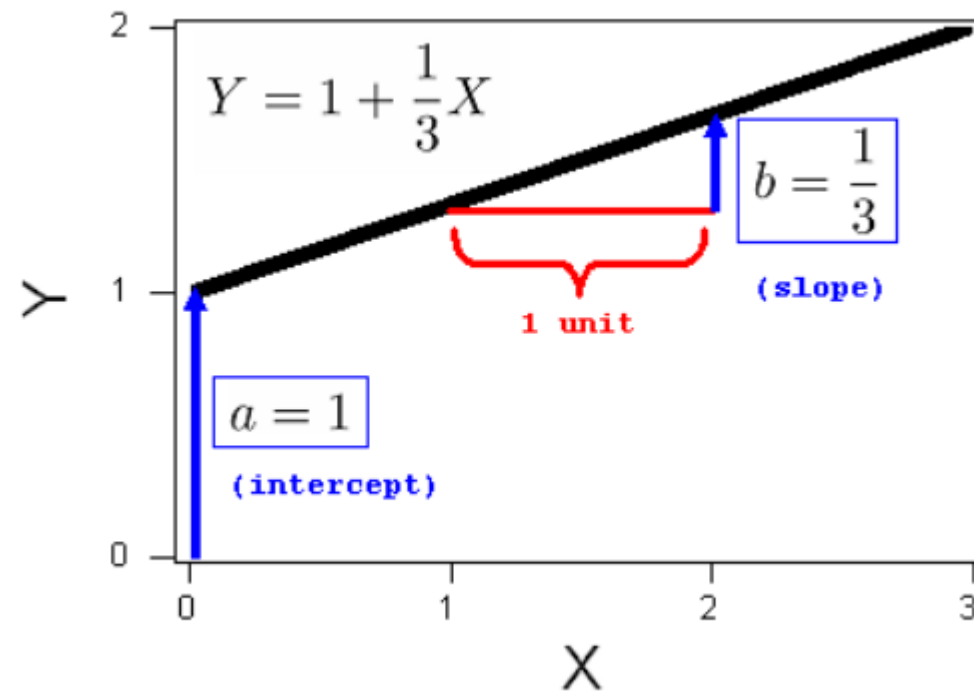
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Note that if either the intercept (a) or the slope (b) is negative, the corresponding blue arrow on the diagram on the right side would point downward.



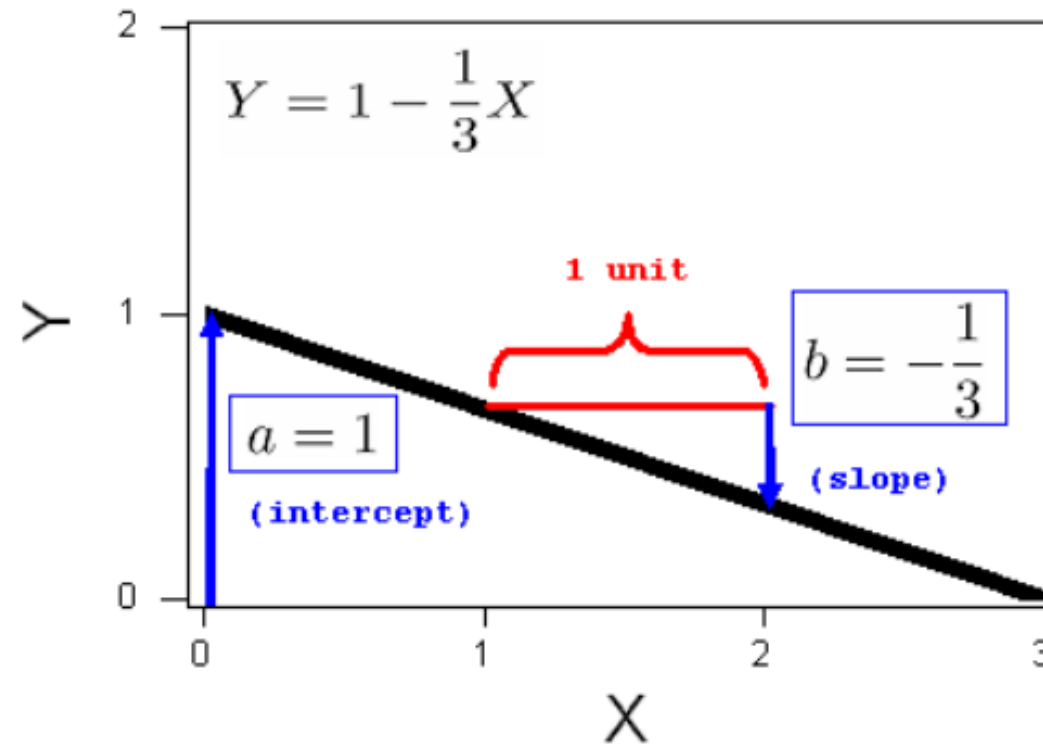
Examining Relationships

Consider the line $Y = 1 + \frac{1}{3}X$. The intercept is 1. The slope is $\frac{1}{3}$, and the graph of this line is, therefore:



Examining Relationships

Consider the line $Y = 1 - \frac{1}{3}X$. The intercept is 1. The slope is $-1/3$, and the graph of this line is, therefore:



Examining Relationships

- All you need to do is calculate the **intercept a** and the **slope b** which is easily done, if we know:
 - $\bar{X}$ – the mean of the explanatory variable's values
 - S_X – the standard deviation of the explanatory variable's values
 - $\bar{Y}$ – the mean of the response variable's values
 - S_Y – the standard deviation of the response variable's values
 - r – the correlation coefficient
- $b = r \left(\frac{S_Y}{S_X} \right)$
 - $a = \bar{Y} - b\bar{X}$

Statistic	Age (X)	Distance (Y)
Mean	51	423
StDev	21.78	82.8
Correlation	-0.793	

- The *slope* of the line is: $b = -0.793 \frac{82.8}{21.78} = -3$. This means that for every 1-unit increase of the explanatory variable, there is, on average, a 3-unit decrease in the response variable. The interpretation *in context* of the slope being -3 is, therefore: For every year a driver gets older, the maximum distance at which he/she can read a sign decreases, *on average*, by 3 feet.
- The *intercept* of the line is: $a = 423 - (-3 \times 51) = 576$ and therefore the *least squares regression line* for this example is:

$$\text{Distance} = 576 + (-3 \times \text{Age})$$



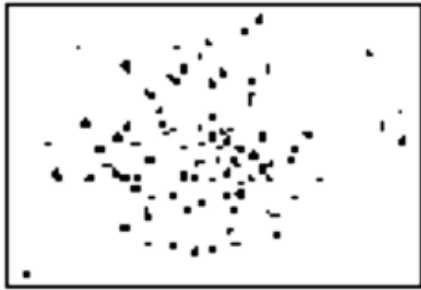
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

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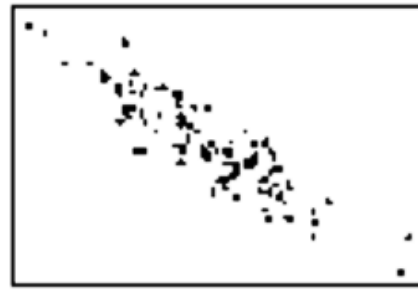
C - Reflect and discuss – Linear Relationship

Question No 06 – Match each scatterplot with the most likely **correlation** value listed below.

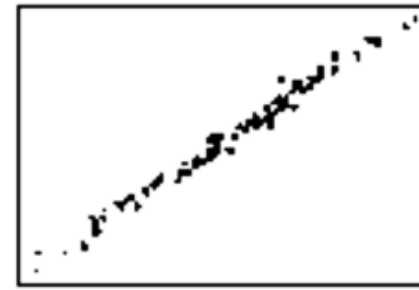
- a. $r = 0.436$.
- b. $r = 0.100$.
- c. $r = -0.897$.
- d. $r = 0.995$.
- e. $r = -0.575$.



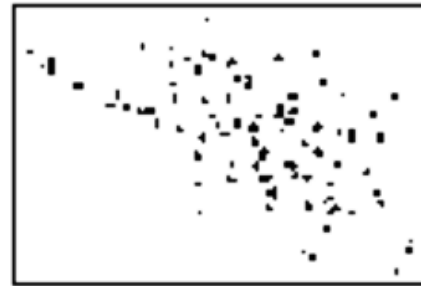
A.



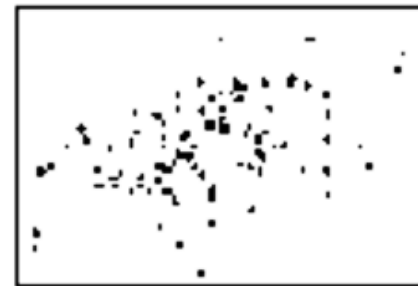
B.



C.



D.



E.

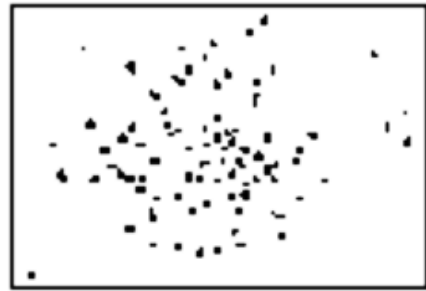
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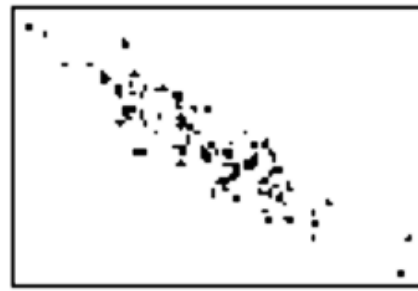
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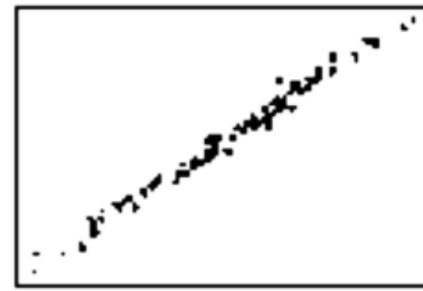
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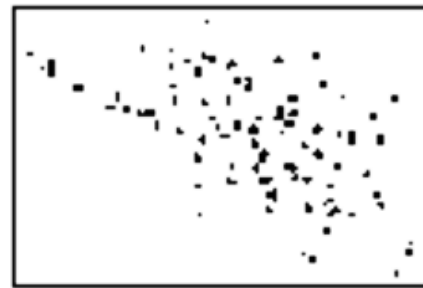
A.



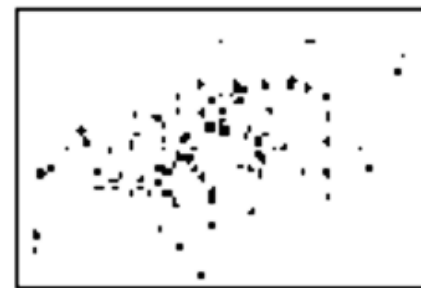
B.



C.



D.



E.

Answer – b

The direction of the relationship is positive and the linear relationship is weak (close to zero).

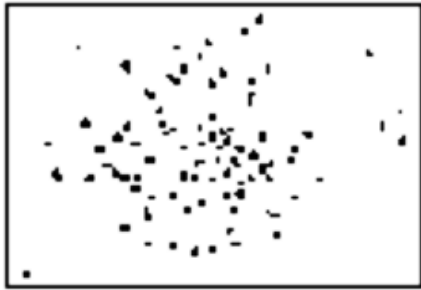
Reading Exercise – No 16 (Module | Statistical Reasoning 2)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

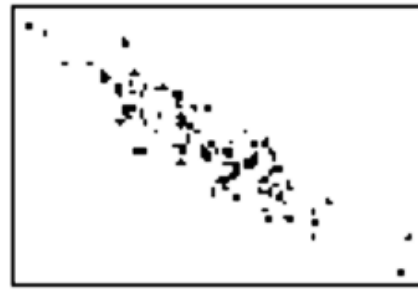
C - Reflect and discuss – Linear Relationship

Question No 06 – Match each scatterplot with the most likely **correlation** value listed below.

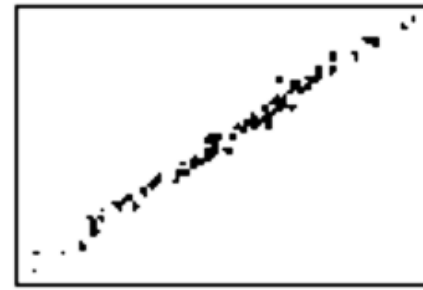
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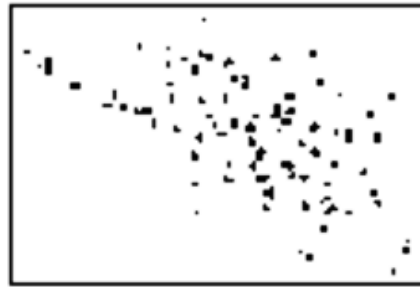
A.



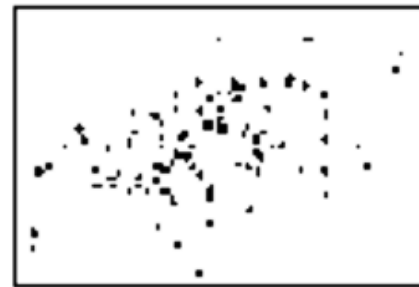
B.



C.



D.



E.

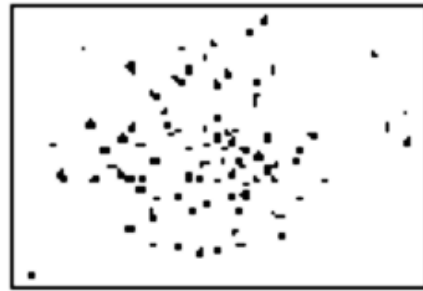
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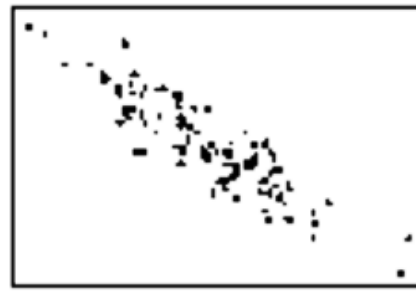
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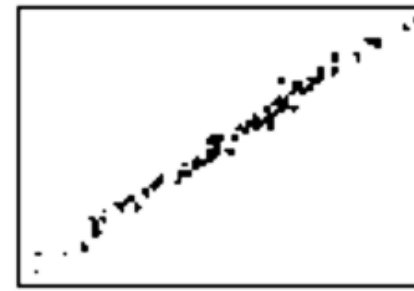
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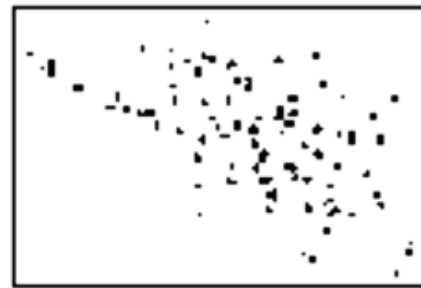
A.



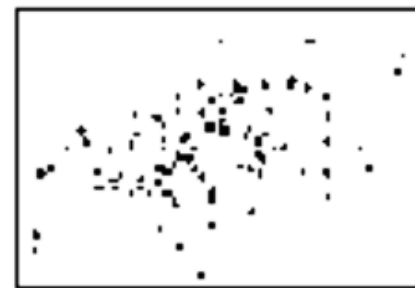
B.



C.



D.



E.

Answer – c

The direction of the relationship is negative, and the linear relationship is quite strong.

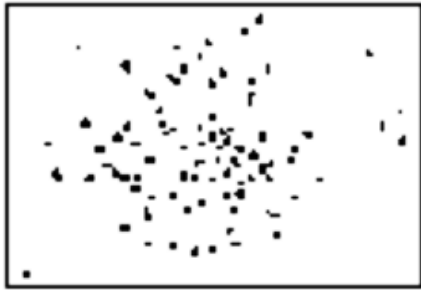
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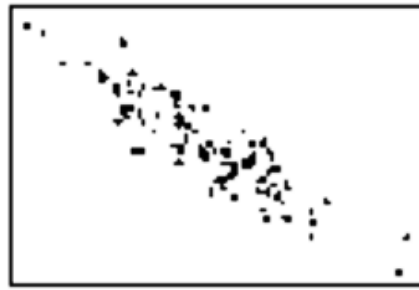
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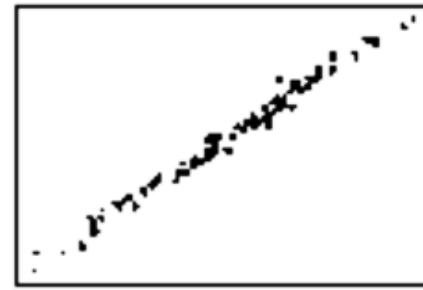
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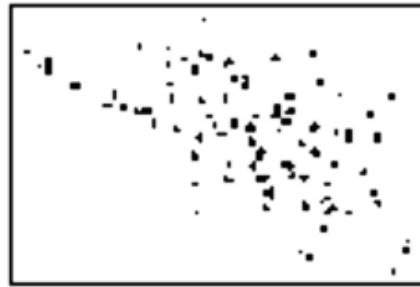
A.



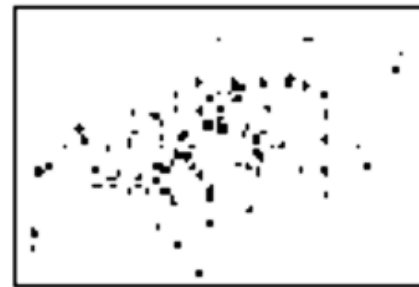
B.



↑ C.



D.



E.

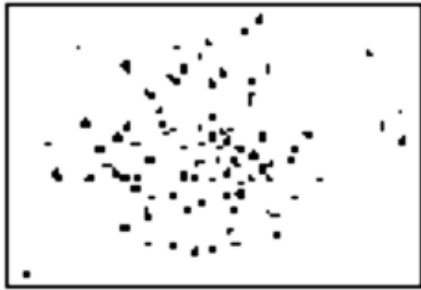
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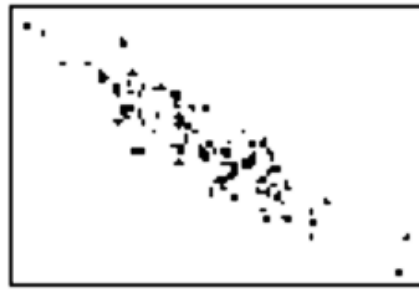
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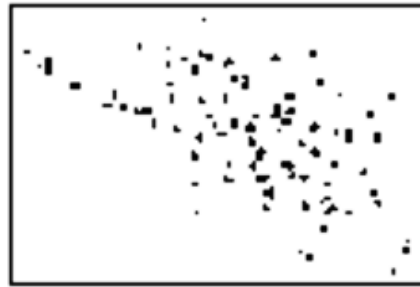
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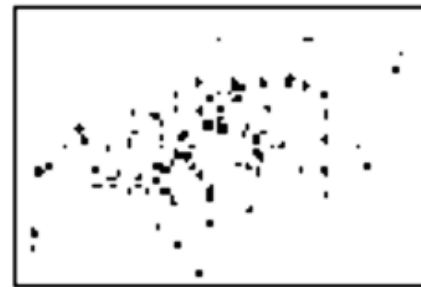
B.



C.



D.



E.

Answer – d

The direction of the relationship is positive, and the linear relationship is quite strong.

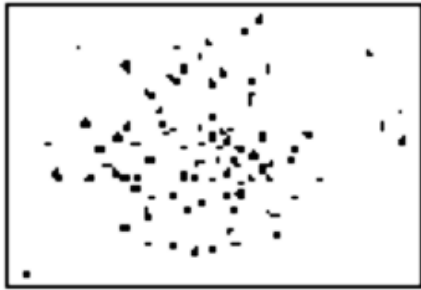
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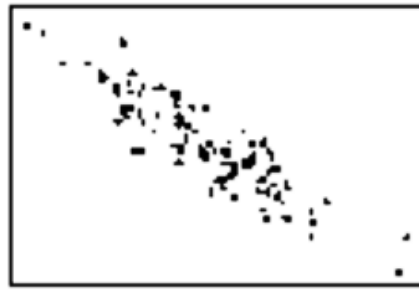
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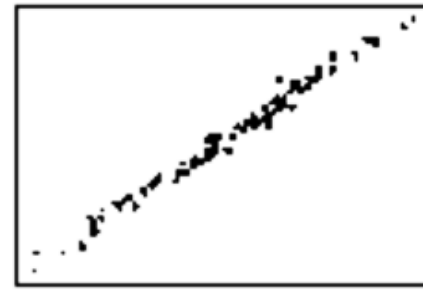
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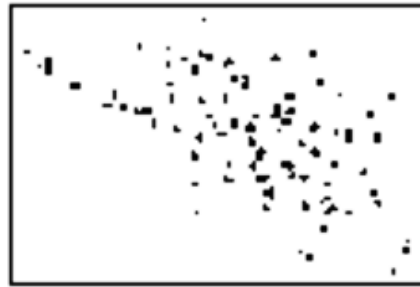
A.



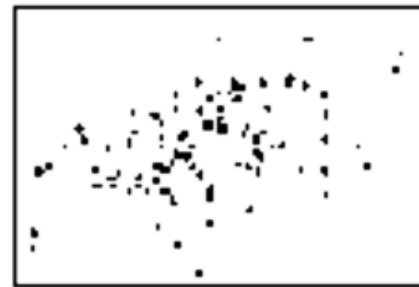
B.



C.



↑ D.



E.

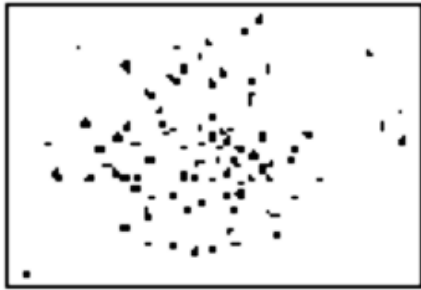
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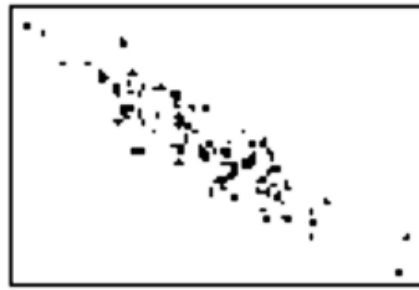
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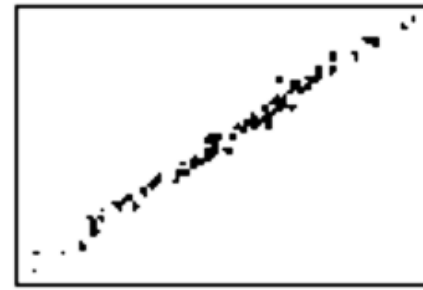
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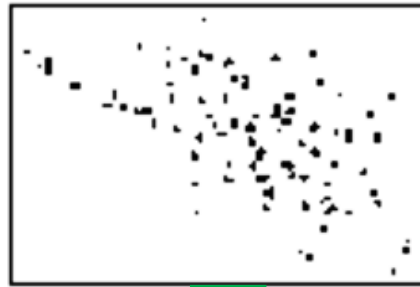
A.



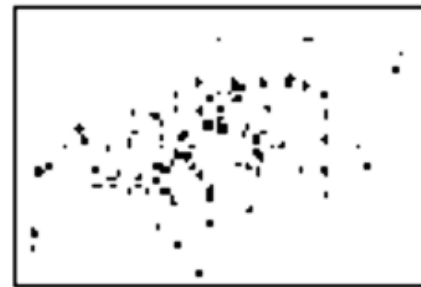
B.



C.



D.



E.

Answer – e

The direction of the relationship is negative, and the linear relationship is only moderately strong.

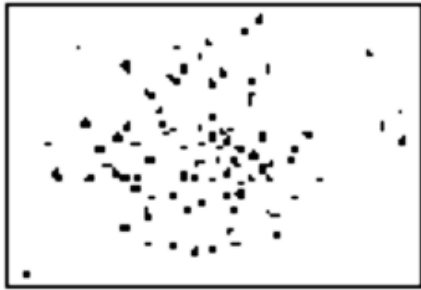
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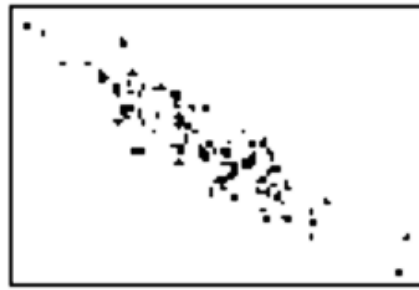
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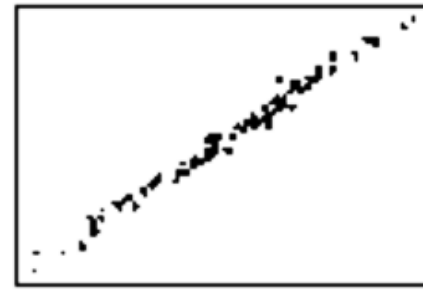
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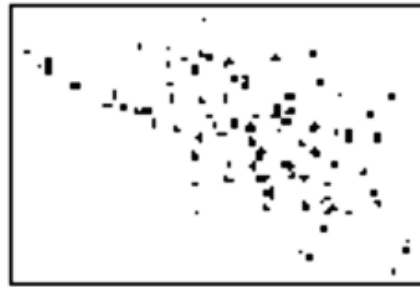
A.



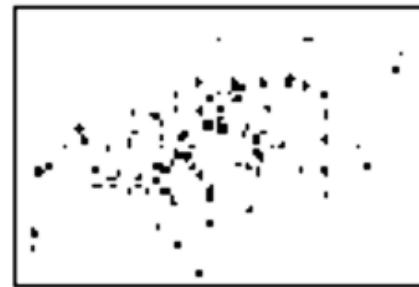
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C.



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E.

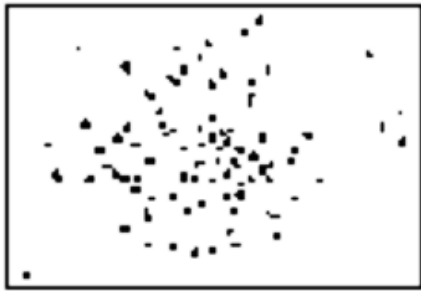
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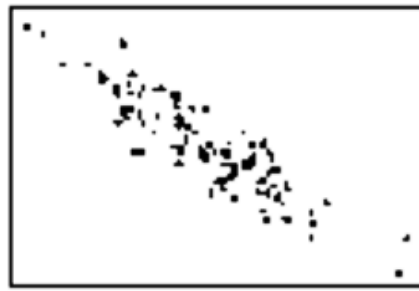
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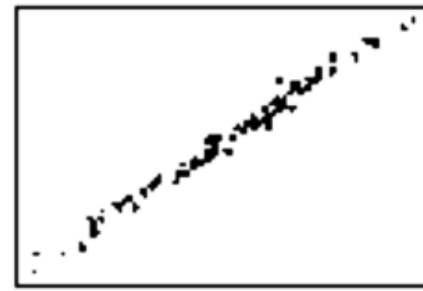
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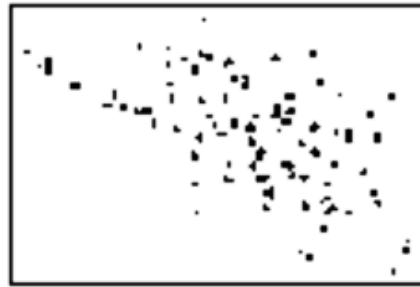
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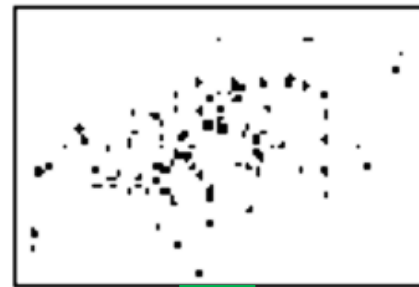
B.



C.



D.

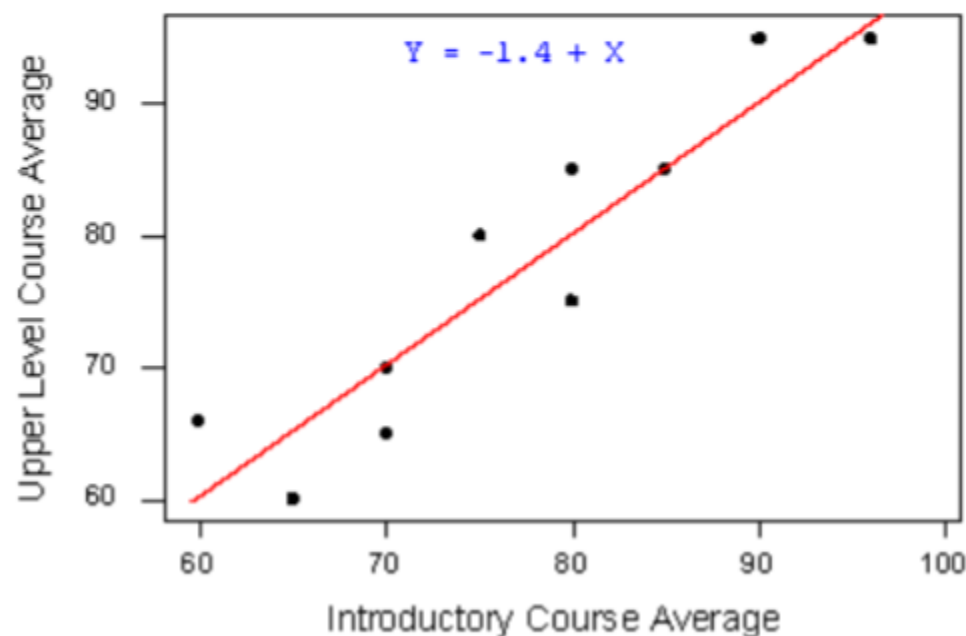


E.

Answer – a

The direction of the relationship is positive, and the linear relationship is only moderately strong.

Background A statistics department is interested in tracking the progress of its students from entry until graduation. As part of the study, the department tabulates the performance of 10 students in an introductory course and in an upper-level course required for graduation. The scatterplot below includes the least squares line (the line that best explains the upper-level course average based on the lower-level course average), and its equation:



Reading Exercise – No 16 (Module | Statistical Reasoning 2)

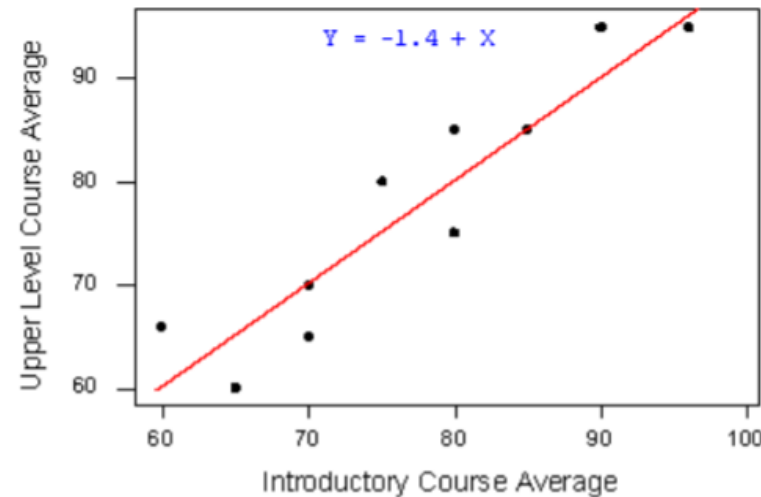
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D - Reflect and discuss – Linear Relationship

Question No 07 – What is the **slope** of the regression?

- A. 1.4
- B. -1.4
- C. 1
- D. -1

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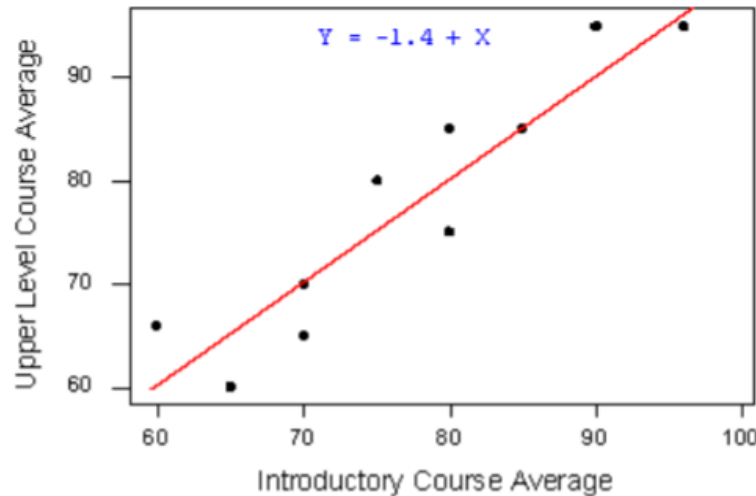
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Answer – c

The equation of the line has the following form: $Y: \text{intercept} + (\text{slope} \times X)$. The number before X is 1 ($1 \times X = X$).

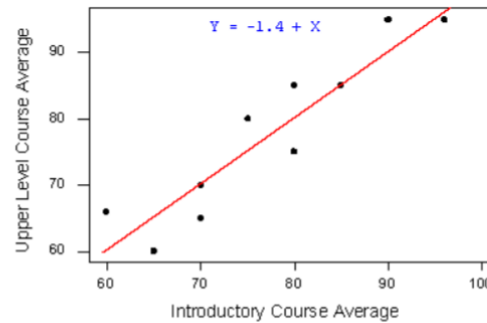
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D - Reflect and discuss – Linear Relationship

Question No 08 – Which of the following is the correct interpretation of the **slope of the regression line?**

Background A statistics department is interested in tracking the progress of its students from entry until graduation. As part of the study, the department tabulates the performance of 10 students in an introductory course and in an upper-level course required for graduation. The scatterplot below includes the least squares line (the line that best explains the upper-level course average based on the lower-level course average), and its equation:



- A. A student who performs 1 point better than another on the upper-level course is likely to perform 1 point worse in the introductory course.
- B. A student who performs 1 point worse than another on the upper-level course is likely to perform 1 point better in the introductory course.
- C. A student who performs 1 point better than another in the introductory course is likely to perform $10 - 1.4 = 8.6$ points better in the upper-level course.
- D. A student who performs 1 point better than another on the introductory course is likely to perform 1 point better in the upper-level course.

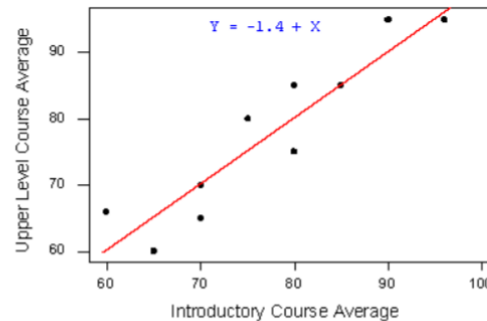
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- D. A student who performs 1 point better than another on the introductory course is likely to perform 1 point better in the upper-level course.**

Answer – d

The slope is the change we would expect in the response variable for an increase of 1 unit in the explanatory variable. A positive slope of 1.0 for the regression line means that every change of 1 unit in the explanatory variable leads us to expect a change of 1 unit in the response variable.

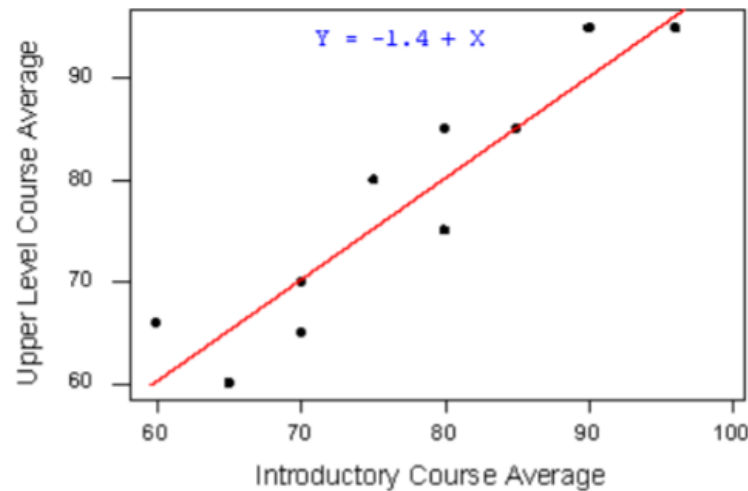
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D - Reflect and discuss – Linear Relationship

Question No 09 – A student scored an average of 90 in the introductory course. What score can we expect for this student in the upper-level course?

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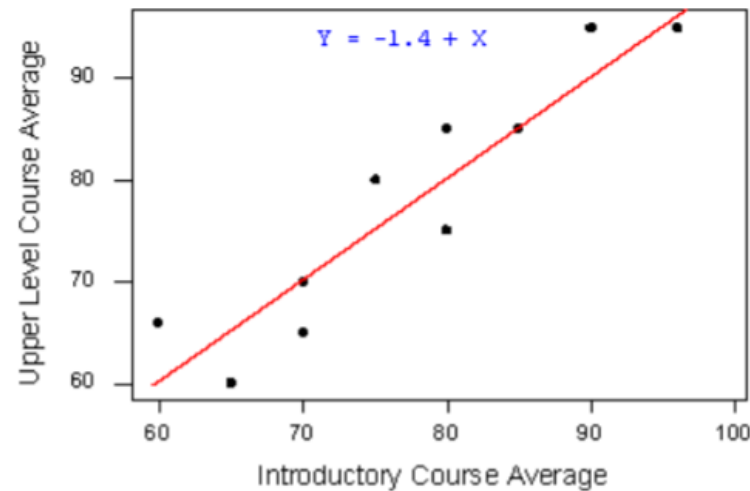
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Answer –

Using the regression line, the correct answer is $-1.4 + 90 = 88.6$